

Chapter 1

Introduction

1.1 The Consultancy Services

The Ministry of Road Transport and Highways, vide letter number NH-12018/03/2012/NG/SARDP-NE Dated 27th December 2013, has issued Letter of Acceptance to RITES Ltd. The Time Period for completion of activities under Terms of Reference is indicated as 9 months.

Performance Guarantee for subject works was submitted by RITES Ltd. on 20th January 2014. The Client officials has requested for clarification on submitted Performance Guarantee on 24th January 2014. The clarifications and amendments to Original Performance Guarantee were submitted by RITES Ltd. on 7th February 2014.

The Contract agreement has been signed on 17th April 2014. As sequel to signing of Contract Agreement, the Client officials have issued notice to consultant for proceeding with works on 19th May 2014. Accordingly consultancy works have been commenced and are in progress.

1.2 Project Roads

Four Project Roads under present scope of works are:

Table 1-1: Project Roads

S. No.	Name of State Road	Length as per ToR (Km)	Actual Length at Site (Km)	Location between (Lat./Long)	
				Start	End
1	Merangkong – Tamlu – Mon Road	100	98.1	26°36'/94°38'	26°43'/95°02'
2	Changtongya – Longleng Road	35	29.5	26°31'/94°40'	26°29'/94°50'
3	Chakabama – Zunheboto Road	128	121.2	25°39'/94°11'	26°02'/94°31'
4	Pfutsero – Phek Road	66	65.3	25°34'/94°17'	25°41'/94°27'
	Total	329	314.1		

1.1.3. Merangkong – Tamlu – Mon (M T M) Road

Project Road starts from Junction with National Highway 61 near Km 209/500 from Kohima in Merangkong Village and terminates at km Stone No. 161 which is start point of Project 'Sewak' of Border Roads Organization (BRO) in Mon District. The Project Road traverses in Mokokchung, Longleng and Mon Districts of Nagaland. The Project Road passes through mountainous to steep terrain.



Start Point (Junction with NH-61) of M T M Road



End Point of M T M Road

1.2.2 Changtongya – Longleng (C L) Road

Project Road starts from Junction with National Highway 61 in Changtongya Village near Km 191 from Kohima. It connects Longleng District Head Quarters with National Highway. The Project Road traverses through Mokokchung and Longleng districts of Nagaland. The Locations of Start and End Points are detailed in photographs below:



Start Point, Junction with NH-61 in Changtongya

End Point in Longleng

During joint site visit held on 15 May 2014 with Nagaland PWD officials at Mokokchung, the physical end point of the project road was fixed near first traffic point from Changtongya. Length of Project Road is 29.530 Km.

1.2.3 Chakabama – Zunheboto (C Z) Road

Chakabama – Zunheboto Road is part of state Highway (approximate 170 km) from Mokokchung – Chakabama, connecting National Highways 61 (in North) and 29 (in South). The state Highway serves as connectivity to state capital, Zunheboto district headquarters and Nagaland University. The Project Road traverses through Phek and Zunheboto District of Nagaland State.

The Project Road starts from Junction with National Highway 29 in Chakabama Village near Km 27 from Kohima. It terminates at Junction after traversing Zunheboto town, near Assam Rifle establishment gate (km 122/250). The Project Road passes through mountainous to steep terrain.



Start Point, Junction with NH-29 in Chakabama

End Point in Zunheboto

1.2.4 Pfutsero – Phek (P P) Road

Project Road starts from Pfutsero, at junction with National Highway 29 near Km 66/500 from Kohima and Ends in Phek at junction in front of Gate of camp area of 'Assam Rifle'. The Project Road would provide 2-lane connectivity to Phek district Headquarters from NH-29 through interior and remote villages.



Start Point, Junction with NH-29 in Pfutsero

End Point in Phek

During Presentation on submitted Preliminary Report, on 8 Oct 2014, it was communicated by MORT&H Officials that following Roads have been declared as New National Highways in State of Nagaland

- **Changtongya - Longleng** - Longching - Mon - Lapa - Tizit - Assam Border as 'NH 702 New'
- Mokokchung - **Zunheboto - Chozuba** - Thaveopheshu - **Phek** - Lanye as 'NH 702 A'.

In view of above, following sections / roads included in preparation of Revised Detailed Project Report would be part of New National Highway System

- **Changtongya - Longleng** Road for Complete Length of 30 km (approx.)
- **Chozuba** (near Km 38/200 on Chakabama - Zunheboto Road from Chakabama) - **Zunheboto** Section of Chakabama Zunheboto Road, Length 84 Km (Approx.)
- **Assam Riffle Check Post** (End Point of Pfutsero - Phek Road) to **Phek Village** Junction, Section of Pfutsero Phek Road, Length 3.5 km (Approx.)

The remaining roads and Sections of Roads would be part of State Highways of Nagaland State, and hence are proposed to be designed in accordance with State Highway/MDR/ODR Standards as detailed below:

- **Merangkong - Tamlu - Mon** Road as **State Highway** as existing road section from Km 47/300 to Mon Town (Km 98/065) is part of State Highway.
- **Chakabama - Chozuba** Section of Chakabama - Zunheboto Road as **State Highway** as it would serve as connection/link between New National Highway 702 A and NH 29 through Cheteba village and terminating at Chakabama.
- **Pfutsero – Phek village** section of Pfutsero – Phek Road as State Highway as it connects Pfutsero (on NH 29) to Phek (on new NH 702 A) through Porba, Sakarba, Khomi and losami villages.

The Key Plan of all Project roads is presented in Figure 1-1. The Location of project roads detailing important features is presented in Figure 1-2. The Plan detailing existing National Highways and Proposed New National Highways in State of Nagaland is presented in Figure 1-3.

1.3 Submissions under Stage 4

The Present report "**Volume VI: Cost Estimate**" is part of submissions under **Stage 4: Final Detailed Project Report (please refer Global Contents at start of the report)**. This is being submitted in accordance with Para 18.6 of Terms of Reference, Page 74 of Contract Agreement.

The Scope of Works for presenting details in Rate Analysis is as under:

This volume will present the contract package wise cost of each item of work as well as a summary of total cost.

The report has been prepared and structured in accordance with stated requirement and is summarized as below.

Chapter 1: Introduction

Estimate of Cost, Estimate of Quantities and respective item rates from Volume V: Rate Analysis have been presented project road wise In following order

- Merangkong – Tamlu – Mon Road
- Changtongya – Longleng Road
- Chakabama – Zunheboto Road
- Pfutsero – Phek Road

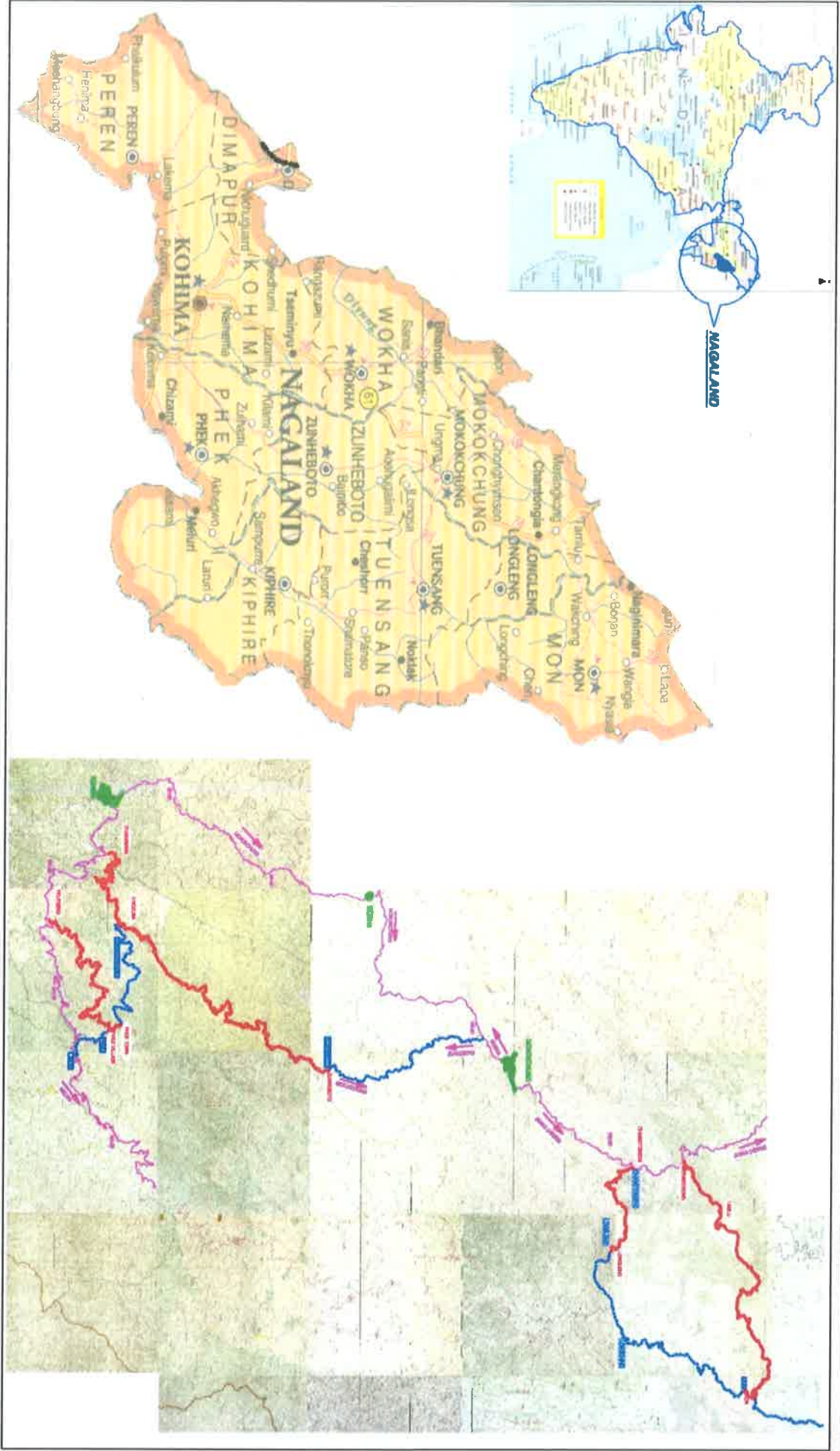
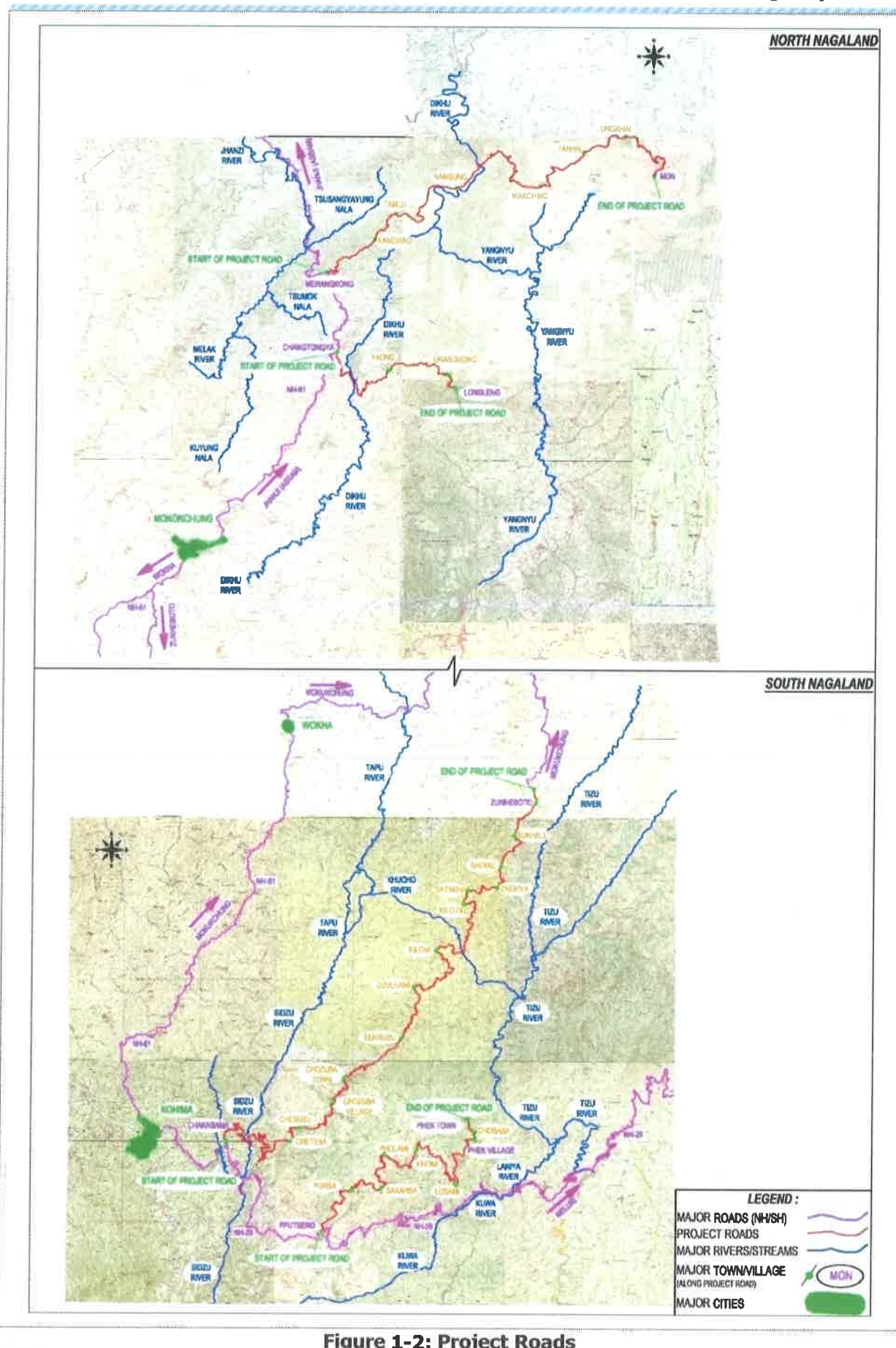


Figure 1-1: Key Plan of Project Roads



Cost Estimate for

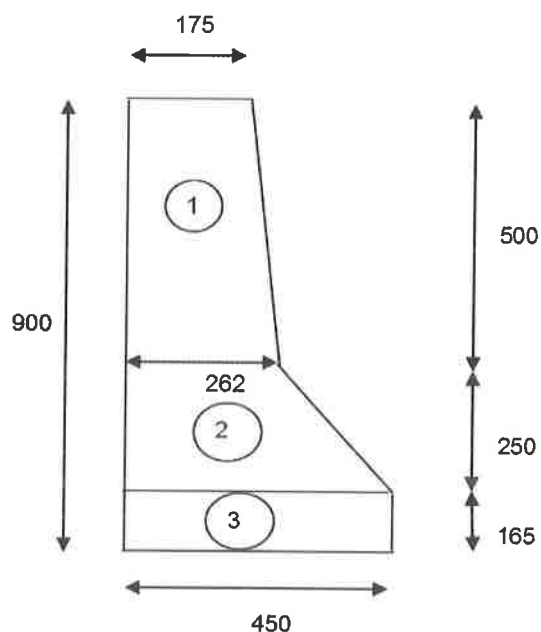
Merangkong – Tamlu – Mon Road

SUMMARY OF QUANTITY CALCULATION FOR BRIDGE AT KM - 79.045			Total Quantity	Unit
<u>Abutment</u> No. of Abutment = 2				
1. Earthwork in excavation for foundation for structures in all types of soil excluding rock :				
Abutment	=	741.855 m ³	1483.71	m ³
2. Filling loose soil in foundations & plinth, including consolidation by watering & ramming in 15cm layers				
Abutment	=	0.000 m ³	0	m ³
Backfilling behind abutments:		819.981 m ³	1639.96	m ³
3. Structural cement concrete for Plain/ Reinforced Concrete in pile cap/open foundation: R.C.C M35 :				
Abutment 11.2 x 13.5 x 1.5	=	226.8 m ³	453.6	m ³
4. Structural cement concrete for Plain/ Reinforced Concrete in substructure, pier, piercap, dirt wall, pedestals, abutment, abutmentcap, retaining wall, return wall, staircase, etc.				
Total :		161.719 m ³	323.44	m ³
RCC M35 in abutment:-				
Dirt Wall	=	12.936 m ³		
Abutment Cap	=	22.453 m ³		
Abutment (489.7-484.5-0.7) x 1.2 x 13.2	=	71.280 m ³		
Return wall	=	52.800 m ³		
RCC M35 in pedestals:-				
= 1.5 x 1.5 x 0.7 x 2	=	2.2500 m ³		
5. Plain cement concrete M 15 leveling course below foundation below approach slabs for bridges :				
Below foundation : 11.4 x 13.7 x 0.15	=	15.618 m ³		
Below approach : 3.5 x 0.15 x 13.2	=	6.930 m ³		
Total :		22.548 m ³	45.096	m ³
6. Structural concrete grade M 30 in Reinforced Cement Concrete in approach slabs excluding reinforcement :				
3.5 x 13.2 x 0.35	=	16.17 m ³	32.34	m ³
7. Providing & fixing in position HYSD reinforcement bars				
a)- Foundation/pile cap, pile, dirt wall & restrainer	=	29.722 T		
b)- Abutment and Return Wall	=	12.408 T		
c)- Abutment Cap	=	13.472 T		
d)- Approach slab	=	0.000 T		
TOTAL =		55.602 T		
Say :		55.60 T	111.20	T
9. Weep Holes :			302.4	Nos
10. Filter media behind abutment			127	m ³

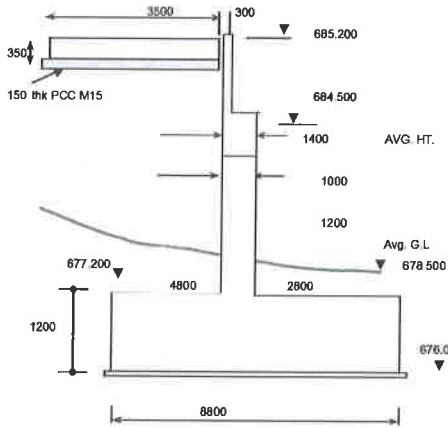
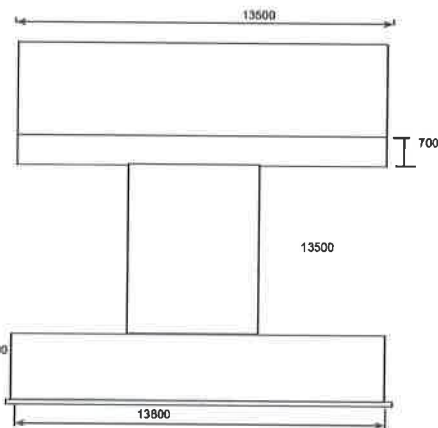
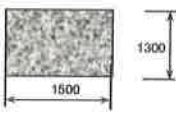
SUMMARY OF QUANTITY CALCULATION FOR BRIDGE AT KM - 79.045

CALCULATION OF QUANTITY FOR CRASH BARRIER AND RAILING

Total length of crash barrier = 50 m
Grade of concrete = M40
Unit weight of concrete (t/m^3) = 2.5

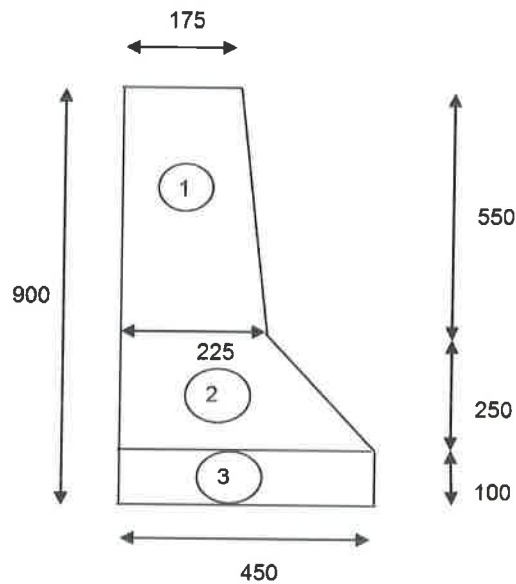


Total volume per m length of crash barrier =	0.350 m^3
Dead weight due to per m length of crash barrier =	0.875 t
Total volume of concrete in crash barrier =	17.5 m^3
Reinforcement quantity =	1.8 T
Total length of MS pipe Railing =	50 m
Hand Railing at the edge of Footpath	50 m

Quantity calculation for bridge at chainage KM 86.795 MTM				Total Quantity	Unit
Abutment					
No. of Abutment = 2  					
1.	Earthwork in excavation for foundation for structures in all types of soil excluding rock : Abutment	=	579.600 m ³	1159.2	m ³
2.	Filling loose soil in foundations & plinth, including consolidation by watering & ramming in 15cm layers Abutment	=	0.000 m ³	0	m ³
	Backfilling behind abutments:	=	714.687 m ³	1429.37	m ³
3.	Structural cement concrete for Plain/ Reinforced Concrete in pile cap/open foundation: R.C.C M35 : Abutment 8.8 x 13.8 x 1.2	=	145.7 m ³	291.456	m ³
4.	Structural cement concrete for Plain/ Reinforced Concrete in substructure, pier, piercap, dirt wall, pedestals, abutment, abutmentcap, retaining wall, return wall, staircase, etc. RCC M35 in abutment:- Dirt Wall Abutment Cap Abutment (684.5-677.2-0.7) x 1.1 x 13.5 Return wall RCC M35 in pedestals:-	Total : = = = = =	156.315 m ³ 2.835 m ³ 13.230 m ³ 98.010 m ³ 42.240 m ³ 0.0000 m ³	312.63	m ³
					
5.	Plain cement concrete M 15 leveling course below foundation below approach slabs for bridges : Below foundation : 9 x 14 x 0.15 Below approach : 3.5 x 0.15 x 13.5	= = Total :	12.600 m ³ 7.088 m ³ 19.688 m ³	39.375	m ³
6.	Structural concrete grade M 30 in Reinforced Cement Concrete in approach slabs excluding reinforcement : 3.5 x 13.5 x 0.35	=	16.64 m ³	33.075	m ³
7.	Providing & fixing in position HYSD reinforcement bars a)- Foundation/pile cap,pile,dirt wall & restrainer b)- Abutment c)- Abutment Cap d)- Approach slab	= = = = TOTAL = Say :	18.443 T 14.025 T 7.938 T 0.000 T 40.406 T 40.41 T	80.81	T
9.	Weep Holes :			175	Nos
10.	Filter media behind abutment			130	m ³
11 a)	Grouting with M15 concrete around toe slab	=	14.532 m ³	29.064	m ³
11 b)	Boulder grouting	=	40.6 m ³	81.2	m ³

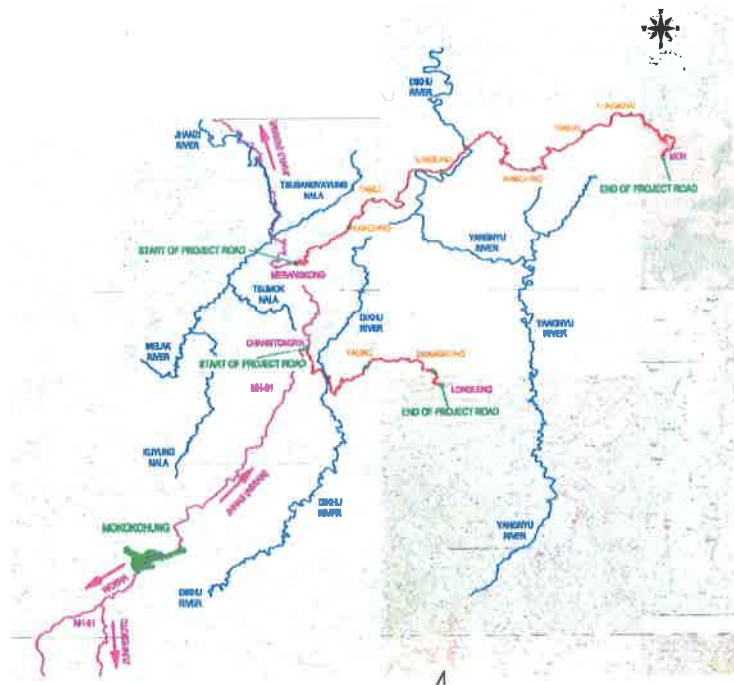
CALCULATION OF QUANTITY OF CRASH BARRIER AND RAILING

Total length of crash barrier = 34 m
Grade of concrete = M40
Unit weight of concrete (t/m^3) = 2.5



Volume of triangle 1 = 0.110 m^3/m
Volume of triangle 2 = 0.084 m^3/m
Volume of triangle 3 = 0.045 m^3/m

Total volume per m length of crash barrier = 0.239 m^3
Dead weight due to per m length of crash barrier = 0.598 t
Total volume of concrete in crash barrier = 8.1 m^3
Reinforcement quantity = 0.8 T
Total length of MS pipe Railing = 34 m
Length of Metal Beam Crash Barrier over Retaining Wall = 20 m
Hand Railing at the edge of Footpath = 34 m



Submitted By



(A Government of India Enterprise)

Highway Division, 4th Floor,
RITES BHAWAN, PLOT No. 1
Sector 29, Gurgaon 122001
Haryana (INDIA)

Document Reference No. 20

RITES/HW/MoRTH-Nagaland/4055-30/2013/FDPR

SUMMARY OF COST

Bill No.	Description	Amount (Rs.)	Figure in Crore
A	Civil Works		
1	SITE CLEARANCE AND DISMANTALING	1,41,27,661	1.41
2	EARTH WORKS	83,15,31,036	83.15
3	GRANULAR SUB-BASE AND BASE COURSES	1,43,40,95,177	143.41
4	BITUMINOUS COURSES AND CONCRETE PAVEMENTS	85,70,42,688	85.70
5	CROSS DRAINAGE WORKS	1,62,55,10,839	162.55
6	BRIDGES / UNDERPASS	47,51,53,196	47.52
7	REPAIR OF EXISTING STRUCTURES	0	0.00
8	DRAINAGE AND PROTECTIVE WORKS	4,62,27,62,595	462.28
9	TRAFFIC SIGNS, MARKINGS AND ROAD APPURTENANCES	23,59,19,826	23.59
10	MISCELLANEOUS	7,26,39,590	7.26
Sub Total for Civil Works (A)		10,16,87,82,607	1016.88
B	Utility Shifting Cost	2,00,00,000	2.00
Sub Total (A+B)		10,18,87,82,607	1018.88
C	Physical Contingencies @5% on (A+B)	50,94,39,130	50.94
Grand Total(A+B+C)		10,69,82,21,738	1069.82
Length of the package Road (km)		86.8	Km
Cost per km on TPC		12.3	Crore

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

RTES Ltd.
Highway Division

**BILL OF QUANTITIES (BOQ)
BILL NO. 1: GENERAL AND SITE CLEARANCE**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
1.01	Clearing and Grubbing of Roadway as per Technical Specification Clause 201.	Ha	153	50,987	78,19,224
1.02	Dismantling of Existing Structures or any Existing Civil Works within Right of Way, as per Technical Specification Clause 202.				
	(i) Brick Masonry	cum	Nil	347	-
	(ii) Stone Masonry	cum	8,495	347	29,47,765
	(iii) Lime/Plain Cement Concrete in all grades	cum	825	509	4,19,925
	(iv) Reinforced / Pre-stressed Cement Concrete in all grades	cum	930	871	8,10,030
	(v) Bituminous courses	cum	2,840	317	9,00,137
	(vi) Non Bituminous courses	cum	Nil	563	-
	(vii) Cement concrete Pavements	cum	Nil	1,654	-
	(viii) Ordinary KM stone/Guard stone/Sign Post	No.	50	249	12,450
	(ix) Hectometre Stone	No.	Nil	50	-
	(x) 5th KM stone	No.	Nil	428	-
	(xi) Kerb stone	m	Nil	16	-
	(xii) All types of Fencing including posts, foundation concrete, back filling of pit etc. complete.	m	Nil	55	-
	(xiii) RCC Railing	m	Nil	253	-
	(xiv) Guard Rails/W Beam Crash Barrier	m	Nil	85	-
	(xv) Brick Pitching / Brick soling	cum	Nil	219	-
	(xvi) Removal of abandoned Telephone / Electric Poles and Lines including excavation and dismantling of foundation concrete and lines under the supervision of concerned department.	m	Nil	192	-
	(xvii) Removing all types of hume pipes and stacking serviceable material with all leads & lifts including earthwork and dismantling of masonry works.		Nil		-
	(a) Up to 600 mm dia	m	Nil	206	-
	(b) Above 600 mm to 900 mm dia	m	1,420	279	3,96,180
	(c) Above 900 mm dia	m	Nil	478	-
	(xviii) Dismantling stone pitching/ dry stone spalls	cum	Nil	235	-
	(xix) Dismantling of CI Water Pipe Line up to 1000 mm dia.	m	Nil	143	-
Carried to Page Collection					1,33,05,711

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

RTES Ltd.
Highway Division

**BILL OF QUANTITIES (BOQ)
BILL NO. 1:GENERAL AND SITE CLEARANCE**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
1.03	Cutting of standing trees as per Technical Specification Clause 201				
	i) Girth from 300mm to 600mm	Each	200	57	11,400
	ii) Girth above 600mm to 900mm	Each	250	102	25,500
	iii) Girth above 900mm to 1800mm	Each	300	199	59,700
	iv) Girth above 1800mm	Each	250	378	94,500
1.04	Removal of Stumps and roots as per Technical Specification Clause 201 and 305.				
	i) Girth from 300mm to 600mm	Each	220	172	37,840
	ii) Girth above 600mm to 900mm	Each	275	306	84,150
	iii) Girth above 900mm to 1800mm	Each	330	597	1,97,010
	iv) Girth above 1800mm	Each	275	1,134	3,11,850
	Carried to Page Collection				8,21,950

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

MITES Ltd.
Highway Division

**BILL OF QUANTITIES (BOQ)
BILL NO. 2: EARTHWORK**

Item No.	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
2.01	Excavation for Roadway including Drains in all types of soils and Ordinary Rocks as per Technical Specifications Clause 301.	cum	23,42,470	235	55,04,80,450
2.02	Excavation for Roadway including Drains in Hard Rock (requiring blasting) as per Technical Specification Clause 301 and 302.	cum	2,06,251	393	8,10,56,643
2.03	Excavation for Roadway including Drains in Hard Rock (using controlled blasting) as per Technical Specification Clause 301 and 302.	cum	55,100	385	2,12,13,500
2.04	Excavation for Roadway including Drains in Hard Rock (blasting prohibited) as per Technical Specification Clause 301.	cum	Nil	515	-
2.05	Construction of Embankment complete as per Technical Specification Clause 305.	cum	4,33,517	233	10,10,09,461
2.06	Construction of Subgrade and Earthen Shoulders complete as per Technical Specification Clause 305.	cum	1,99,796	268	5,35,45,328
2.07	Preparation of Cut Formation as Subgrade satisfying designed CBR at 97% MDD complete as per Technical Specification Clause 301 and 305.	cum	2,91,669	66	1,92,50,154
2.08	Preparation of Subgrade in Rocky Formation as per Technical Specification Clause 301.	sqm	16,050	310	49,75,500
Carried to Page Collection					83,15,31,036

**BILL OF QUANTITIES (BOQ)
BILL NO.5: CROSS DRAINAGE STRUCTURES (CULVERTS)**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
5.01	Excavation for Foundation of Structure, as per drawings and Technical Specification, in				
	(a) All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	35,524	235	83,48,140
	(b) Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	Nil	393	-
	(c) Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	Nil	385	-
	(d) Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	82,888	515	4,26,87,320
5.02	Plain Cement Concrete (M 15 Grade) for Levelling Course below Headwall, Abutment, Wing wall, Around Foundation in rock, Approach slab, Chutes, Catch Pits etc. complete as per drawings and Technical Specification Section 1700, 2100, 2200, 2500 and 2700.	cum	3,568	10,913	3,89,37,584
5.03	RCC/PCC for open foundations complete excluding reinforcement as per drawings and Technical Specification Section 1700 and 2100				
	(a) M 25 Grade	cum	Nil	13,123	-
	(b) M 30 Grade	cum	4,494	13,132	5,90,15,208
5.04	RCC / PCC for Substructure, Wing wall, Retaining wall, Box Cell Structures etc. excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2200.				
	(a) M 25 Grade	cum	Nil	13,914	-
	(b) M 30 Grade	cum	22,718	13,958	31,70,97,844
5.05	Coursed Rubble Masonry in 1:3 Cement Mortar for Foundation and Substructure complete as per Drawing and Technical Specification Clause 1400, 2100 and 2200.	cum	Nil	10,671	-
5.06	RCC (M 30 Grade) for Slab Culvert Superstructure excluding reinforcement complete as per drawings and Technical specifications Section 1700 and 2300.	cum	Nil	15,121	-
5.07	Steel Reinforcement (Fe 500D) for Foundations, Substructures and superstructures etc. complete as per drawings and Technical Specification Section 1600.	MT	3,057	91,836	28,07,42,652
5.08	Approach Slab (RCC M 30 Grade) including reinforcement complete as per drawings and Technical Specification Section 2700.	cum	3,938	17,586	6,92,53,668
5.09	RCC Crash Barrier (M 40 Grade) excluding cost of reinforcement complete as per drawings and Technical Specification Section 1700 and 2700	cum	Nil	15,431	-
5.10	RCC (M 30 Grade) comprising cast in-situ Post and Precast Hand rails including reinforcement complete as per drawings and Technical Specification Section 1600, 1700 and 2700.	Rm	Nil	3,021	-
	Carried to Page Collection				81,60,82,416

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

RTES Ltd.
Highway Division

**BILL OF QUANTITIES (BOQ)
BILL NO.5: CROSS DRAINAGE STRUCTURES (CULVERTS)**

Item No	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
5.11	RCC Pipes, having diameter (in mm) as detailed below, for culverts complete as per drawings and Technical Specification Section 2900				
	(a) 900	m	Nil	17,011	-
	(b) 1000	m	Nil	18,901	-
	(c) 1200	m	3,813	22,075	8,41,71,975
	(d) 1500	m	Nil	27,594	-
5.12	Filter Media behind Earth Retaining Structures (abutment, splayed wing wall, retaining wall, return wall etc.) complete as per Technical Specification Section 2500	cum	12,940	3,539	4,57,94,660
5.13	Back filling behind Earth Retaining Structures (abutments, splayed wing walls, return wall, box cell structures etc.) Complete as per drawings and Technical Specification Clause 305.	cum	70,643	3,670	25,92,59,810
5.14	Tar Paper Bearing complete as per drawings Technical Specification Section 2000.	sqm	Nil	140	-
5.15	Expansion Joint (Filler Joint) complete as per Drawings and Technical Specification Section 2600.	m	Nil	363	-
5.16	Expansion Joint (Asphaltic Plug Joint Type) Complete as per Drawings and Technical Specification Section 2600	m	3,215	1,715	55,13,725
5.17	Drainage spouts complete as per drawings and Technical Specifications Section 2700.	nos	596	4,229	25,20,484
5.18	Filter Material underneath Boulder / Concrete Block Pitching complete as per drawing and Technical Specification Section 2500.	cum	Nil	3,743	-
5.19	Boulder / Concrete Block Pitching on Embankment Slopes, In front of Abutment and In Launching Apron complete as per drawings and Technical Specification Section 2500.	cum	Nil	4,283	-
5.20	Boulder / Concrete Block Pitching in hand woven wire Crates on Embankment slopes in front of Abutment and in launching apron complete as per drawings and Technical Specification Section 2500	cum	20,507	6,295	12,90,91,565
5.21	Mechanically Woven Revetment / Gabion mattress filled with Boulder / Concrete block for erosion control, river training and protection works complete as per drawings and Technical Specification	cum	Nil	6,325	-
5.22	Plain Cement Concrete (M 15 Grade) for Cement Concrete Bedding for Protection works complete as per drawings and Technical Specification Section 1700 and 2500.	cum	3,988	11,349	4,52,59,812
5.23	Flat Stone Apron for Bed Protection Works complete as per drawings and Technical Specification Section 1300, 1400 and 2500	cum	Nil	10,005	-
5.24	Lettering and Number of Culverts as per drawings and Additional Technical Specification A-3.	nos	21,600	1	15,120
5.25	Painting with water based cement paint on concrete surface as per additional Technical	sqm	2,346	186	4,36,356
5.26	Weepholes (AC/PVC/HDPE) in earth retaining structure,complete as per drawings and Technical Specification.	nos.	17,280	633	1,09,38,240
5.27	RCC/PCC for rigid flooring/buffe pier,blocks,chutes etc.excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2200	cum	16,222	13,958	22,64,26,676
Carried to Page Collection					80,94,28,423

BILL OF QUANTITIES
BILL NO. 6: NEW BRIDGES

Item	Description	Unit	Quantity	Unit Rate	Amount (Rupees)
6.10	Bearings, of following Type, as per drawings and Technical Specification Section 2000				
a	Tar Paper Bearings	sqm	27	140	3,780
b	Elastomeric Bearings	cu.cm	Nil	7	-
c	POT cum PTEE Bearings		Nil		-
i	Fixed Bearings, having Capacity of		Nil		-
	925 T ± 25 T	nos	Nil	1,61,823	-
	525 T ± 25 T	nos	Nil	91,845	-
	325 T ± 25 T	nos	Nil	29,591	-
	275 T ± 25 T	nos	Nil	25,039	-
ii	Free / Guided Bearings, havirg Capacity of		Nil		-
	925 T ± 25 T	nos	Nil	2,35,298	-
	825 T ± 25 T	nos	Nil	2,09,861	-
	725 T ± 25 T	nos	Nil	1,84,423	-
	675 T ± 25 T	nos	Nil	1,71,704	-
	575 T ± 25 T	nos	Nil	1,46,267	-
	525 T ± 25 T	nos	Nil	1,33,548	-
	325 T ± 25 T	nos	Nil	41,832	-
	275 T ± 25 T	nos	88	35,396	31,14,848
iii	Pin Bearings, having Capacity of		Nil		-
	50 T ± 12.5 T	nos	Nil	16,783	-
	75 T ± 12.5 T	nos	Nil	25,174	-
	100 T ± 12.5 T	nos	Nil	33,565	-
	125 T ± 12.5 T	nos	Nil	50,682	-
	150 T ± 12.5 T	nos	2	71,289	1,42,578
iv	Metallic Guide Bearings, having Capacity of		Nil		-
	25 T ± 12.5 T	nos	Nil	8,237	-
	50 T ± 12.5 T	nos	Nil	16,475	-
	75 T ± 12.5 T	nos	Nil	24,712	-
	100 T ± 12.5 T	nos	2	32,949	65,898
6.11	Elastomeric Stopper Pad for Seismic Restrainer as per drawings and Technical Specification 2000	cu.cm	Nil	7	-
Carried to Page Collection					33,27,104

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

RTIES Ltd.
Highway Division

BILL OF QUANTITIES
BILL NO. 6: NEW BRIDGES

Item	Description	Unit	Quantity	Unit Rate	Amount (Rupees)
6.12	Expansion Joints, of following Type as per drawings and Technical Specification Section 2600				
a	Filler	m	27	363	9,801
b	Asphaltic Plug	m	Nil	1,715	-
c	Strip Seal Type capable of taking		Nil		-
i	40 mm movement	m	100	6,904	6,90,400
ii	80 mm movement	m	48	46,353	22,24,944
iii	160 mm movement	m	60	75,968	45,58,080
6.13	Helical Void Former for Voided Concrete Slabs including 20 mm diameter tie rod and seating joints as per drawings and Technical Specification Section 2300 and Additional Technical Specification A	m	192	2,466	4,73,472
6.14	Drainage Spouts as per Technical Specification Section 2700	nos	250	4,229	10,57,250
6.15	Approach Slab (RCC, M 30 Grade) including reinforcement as per drawings and Technical Specification Section 2700	cum	187	17,586	32,88,582
6.16	RCC (M 30 Grade) comprising cast in-situ Post and Precast Hand rails including reinforcement complete as per drawings and Technical Specification Section 1600, 1700 and 2700.	Rm	614	3,021	18,54,894
6.17	RCC Crash Barrier (M 40 Grade) excluding cost of reinforcement complete as per drawings and Technical Specification Section 1700 and 2700	cum	173	16,431	28,42,602
6.18	Filling of Selected imported granular materials below structure as per drawings and Technical Specification Clause 305, 1000 and 2900	Cum	Nil	3,670	-
6.19	Weep holes (AC/PVC/HDPE) in Earth Retaining Structures (Abutments, Splayed Wing wall, Median Wall, Retaining Wall etc.) complete as per drawings and Technical Specification Section 2700.	nos	1,943	633	12,29,919
6.20	Back filling behind Earth Retaining Structures (abutments, splayed wing walls, return wall, box cell structures etc.) Complete as per drawings and Technical Specification Clause 305.	cum	7,887	3,670	2,89,45,290
6.21	Filter Media behind Earth Retaining Structures (abutment, splayed wing wall, retaining wall, return wall etc.) complete as per Technical Specification Section 2500	cum	772	3,539	27,32,108
6.22	Boulder / Concrete Block Pitching on Embankment Slopes, In front of Abutment and In Launching Apron complete as per drawings and Technical Specification Section 2500.	cum	Nil	4,283	-
6.23	Boulder / Concrete Block Pitching in hand woven Wire Crates on Embankment Slopes in front of Abutment and in launching apron complete as per drawings and Technical Specification Section	cum	Nil	6,295	-
6.24	Filter Material underneath Boulder / Concrete Block Pitching complete as per drawing and Technical Specification Section 2500.	cum	Nil	3,743	-
Carried to Page Collection					4,99,07,342

BILL OF QUANTITIES (BOQ)
BILL NO. 9: Traffic sign Marking and other Appurtenances

Item No.	Description	Unit	Quantity	Unit Rate	Amount (Rs.)
9.01	Ground Mounted Traffic (Cautionary, mandatory and information) signs as per Technical Specification Clause 801.				
	(i) 600 mm (Equilateral Triangle)	Each	761	5,750	43,75,750
	(ii) 600 mm (Circular)	Each	144	7,451	10,72,944
	(iii) 600 mm x 450 mm (Rectangular)	Each	98	7,276	7,13,048
	(iv) 600 mm (Octagonal)	Each	79	7,651	6,04,429
	(v) 600 mm x 500 mm (Chevron Signs)	Each	2,928	8,084	2,36,71,253
9.02	Ground Mounted Traffic (Direction and Place Identification) signs complete as per Technical Specification Clause 801				
	(i) up to 0.9 sqm	Each	106	17,443	18,48,958
	(ii) more than 0.9 sqm	Each	3	30,362	91,086
9.03	Overhead/Cantilever Signs complete as per Technical Specification Clause 802				
	i) Structure for Overhead Sign	MT	5	2,48,370	12,41,850
	ii) Aluminum Sheet	sqm	40	1,340	53,600
	iii) Earthwork in excavation of foundation for structure, complete as per drawings, Technical specifications section 300 and 2900.	cum	50	235	11,750
	iv) Plain Cement Concrete (M 15 Grade) for Levelling Course below Headwall, Abutment, Wing wall, Around foundation in rock, Approach slab, Chutes etc. complete as per drawings and Technical Specification Section 1700, 2100, 2200, 2500 & 2700	cum	3	10,913	32,739
	v) RCC M25 for open foundations complete excluding reinforcement as per drawings and Technical Specification Section 1700 and 2100	cum	16	13,123	2,09,968
	vi) Steel Reinforcement Fe 500D in Foundation, Substructures Superstructure etc. complete as per drawings and Technical Specification Section 1600	MT	2	91,836	1,83,672
9.04	Road Markings Complete as per Technical Specifications Clause 803				
	a) Centre line / Edge / Lane / any other marking	sqm	29,845	2,802	8,36,26,645
	b) Directional Arrows / Lettering	No.	60	2,410	1,44,583
9.05	Bi-Directional Reflective Pavement Markers (Road Studs) Complete as per Technical Specification Clause 804.	No.	43,420	1,529	6,63,89,180
Carried to Page Collection					18,42,71,456

**ABSTRACT OF QUANTITIES
BILL NO. 1: GENERAL AND SITE CLEARANCE**

Item No	Description	Unit	No	Length (m)	Breadth h (m)	Height / Depth	Total Quantity
1.C1	Clearing and Grubbing of Roadway as per Technical Specification Clause 201.	Ha	Plan Area of Project	Improvement			153.36
1.C2	Dismantling of Existing Structures or any Existing Civil Works within Right of Way, as per Technical Specification Clause 202.						
	(i) Brick Masonry	cum					
	(ii) Stone Masonry	cum		Lump Sum			8,495
	(iii) Lime/Plain Cement Concrete in all grades	cum		Lump Sum			825
	(iv) Reinforced / Pre-stressed Cement Concrete in all grades	cum		Lump Sum			930
	(v) Bituminous courses	cum	1	40,565	3.5	0.020	2,840
	(vi) Non Bituminous courses	cum					
	(vii) Cement concrete Pavements	cum					
	(viii) Ordinary KM stone/Guard stone/Sign Post	No.					50
	(ix) Hectometre Stone	No.					
	(x) 5th KM stone	No.					
	(xi) Kerb stone	m					
	(xii) All types of Fencing including posts, foundation concrete, back filling of pit etc. complete.	m					
	(xiii) RCC Railing	m					
	(xiv) Guard Rails/W Beam Crash Barrier	m					
	(xv) Brick Pitching / Brick soling	cum					
	(xvi) Removal of abandoned Telephone / Electric Poles and Lines including excavation and	m					
	(xvii) Removing all types of hume pipes and stacking serviceable material with all leads & lifts						
	(a) Up to 600 mm dia	m					
	(b) Above 600 mm to 900 mm dia	m		Lump Sum			1,420
	(c) Above 900 mm dia	m					
	(xviii) Dismantling stone pitching/ dry stone spalls	cum					
	(xix) Dismantling of CI Water Pipe Line up to 1000 mm dia.	m					

ABSTRACT OF QUANTITIES
BILL NO.5: CROSS DRAINAGE STRUCTURES (CULVERTS)

Item No	Description	Unit	No.	Length (m)	Breadth (m)	Height / Depth(m)	Quantity
5.01	Excavation for Foundation of Structure, as per drawings and Technical Specification, in (a) All types of soils and Ordinary Rocks as per Technical specification Clause 304. (b) Hard Rock (requiring blasting) as per Technical specification Clause 304. (c) Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304. (d) Hard Rock (blasting prohibited) as per Technical Specification Clause 302 and 304.	cum	As per annexure 5.1				35,524
5.02	Plain Cement Concrete (M 15 Grade) for Levelling Course below Headwall, Abutment, Wing wall, Around foundation in rock, Approach slab, Chutes, Catch pits etc. complete as per drawings and Technical Specification Section 1700, 2100, 2200, 2500 and 2700.	cum	As per annexure 5.1				82,888
5.03	RCC/PCC for open foundations complete excluding reinforcement as per drawings and Technical Specification Section 1700 and 2100	cum	As per annexure 5.1				3,568
5.04	(a) M 25 Grade (b) M 30 Grade RCC / PCC for Substructure, Wing wall, Retaining wall, Box Cell Structures etc. excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2200.	cum	As per annexure 5.1				4,494
5.05	Coursed Rubble Masonry in 1:3 Cement Mortar for Foundation and Substructure complete as per Drawing and Technical Specification Clause 1400, 2100 and 2200.	cum					
5.06	RCC (M 30 Grade) for Slab Culvert Superstructure excluding reinforcement complete as per drawings and Technical specifications Section 1700 and 2300.	cum	As per annexure 5.1				22,718
5.07	Steel Reinforcement (Fe 500D) for Foundations, Substructures and superstructures etc. complete as per drawings and Technical Specification Section 1600.	cum					
5.08	Approach Slab (RCC M 30 Grade) including reinforcement complete as per drawings and Technical Specification Section 2700.	MT	As per annexure 5.1				
5.09	RCC Crash Barrier (M 40 Grade) excluding cost of reinforcement complete as per drawings and Technical Specification Section 1700 and 2700	cum	As per annexure 5.1				3,057
5.10	RCC (M 30 Grade) comprising cast in-situ Post and Precast Hand rails including reinforcement complete as per drawings and Technical Specification Section 1600, 1700 and 2700.	cum					3,938
5.11	RCC Pipes, having diameter (in mm) as detailed below, for culverts complete as per drawings and Technical Specification Section 2900	Rm					
	(a) 900						
	(b) 1000						
	(c) 1200						
	(d) 1500						
		m					
		m					
		m					
		m	As per annexure 5.1				3,813

Preparation of Revised DPR and Verification of Executed Quantities/Items for widening to 2-lane of 4 roads in state of Nagaland under SARDP-NE

ABSTRACT OF QUANTITIES
BILL NO. 6: BRIDGES

Item	Description	Unit	No.	Length (m)	Breadth (m)	Height / Depth(m)	Quantity
6.01	Excavation for Foundation of Structure, as per drawings and Technical Specification, in						
a	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	As per annexure 6.1				4,939
b	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum					-
c	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum					-
d	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	As per annexure 6.1				6,739
6.02	PCC (M 15 Grade) in Levelling course below Foundation, Pile Cap, Approach slab, Annular filling around foundation in rock, etc. complete as per drawings and Technical Specification Section 1700, 2100, 2500 and 2700.	cum	As per annexure 6.1				367
6.03	RCC / PCC in open foundations excluding reinforcement complete as per drawings and Technical Specification Sections 1700 and 2100.						
a	M 30 Grade	cum					-
b	M 35 Grade	cum	As per annexure 6.1				4,495
6.04	RCC / PCC in Substructure, Wing Wall, retaining wall, Box Cell Structures, Seismic restrainer etc. excluding reinforcement, in following grades, complete as per Drawings and Technical Specification Section 1700 and 2200						
a	M 25	cum					-
b	M 30	cum					-
c	M 35	cum	As per annexure 6.1				3,644
6.05	RCC in Superstructure excluding reinforcement complete as per drawings and Technical Specification Section 1700 and 2300.						
a	Solid Slab (M 30 Grade)	cum	As per annexure 6.1				98
b	Deck Slab for Composite Girder (M 35 Grade)	cum	As per annexure 6.1				570
c	Voided Slab (M 35 Grade)	cum	As per annexure 6.1				370
6.06	Steel Reinforcement Fe 500D in Foundation, Substructures Superstructure etc. complete as per drawings and Technical Specification Section 1600 .	MT	As per annexure 6.1				1,383
6.07	Steel Girder for Steel Composite Superstructure including railing and fixing of girder with Bearing complete as per drawings and Technical Specification 1000 and 1900.	MT	As per annexure 6.1				717
6.08	Bituminous (Type 2) Wearing Coat as per drawings and Technical Specification Section 2700.	sqm	As per annexure 6.1				2,290
6.09	40 thk. PCC (M25) finished with 15 thk. Plaster (1:3) complete as per drawings and Technical Specification.	cum	As per annexure 6.1				44

ABSTRACT OF QUANTITIES
BILL NO. 6: BRIDGES

Item	Description	Unit	No.	Length (m)	Breadth (m)	Height / Depth(m)	Quantity
6.11	Elastomeric Stopper Pad for Seismic Restrainer as per drawings and Technical Specification 2000	cu.cm					-
6.12	Expansion Joints, of following Type as per drawings and Technical Specification Section 2600						
a	Filler	m	As per annexure 6.1				27
b	Asphaltic Plug	m					-
c	Strip Seal Type capable of taking						
i	40 mm movement	m	As per annexure 6.1				100
ii	80 mm movement	m	As per annexure 6.1				48
iii	160 mm movement	m	As per annexure 6.1				60
6.13	Helical Void Former for Voided Concrete Slabs including 20 mm diameter tie rod and seating joints as per drawings and Technical Specification Section 2300 and Additional Technical Specification A	m	As per annexure 6.1				
6.14	Drainage Spouts as per Technical Specification Section 2700	nos					192
6.15	Approach Slab (RCC, M 30 Grade) including reinforcement as per drawings and Technical Specification Section 2700	cum	As per annexure 6.1				250
6.16	RCC (M 30 Grade) comprising cast in-situ Post and Precast Hand rails including reinforcement complete as per drawings and Technical Specification Section 1600, 1700 and 2700.	Rm	As per annexure 6.1				614
6.17	RCC Crash Barrier (M 40 Grade) excluding cost of reinforcement complete as per drawings and Technical Specification Section 1700 and 2700	cum	As per annexure 6.1				173
6.18	Filling of Selected imported granular materials below structure as per drawings and Technical Specification Clause 305, 1000 and 2900	Cum					-
6.19	Weep holes (AC/PVC/HDPE) in Earth Retaining Structures (Abutments, Splayed Wing wall, Median Wall, Retaining Wall etc.) complete as per drawings and Technical Specification Section 2700.	nos	As per annexure 6.1				1,943
6.20	Back filling behind Earth Retaining Structures (abutments, splayed wing walls, return wall, box cell structures etc.) Complete as per drawings and Technical Specification Clause 305.	cum	As per annexure 6.1				7,887
6.21	Filter Media behind Earth Retaining Structures (abutment, splayed wing wall, retaining wall, return wall etc.) complete as per Technical Specification Section 2500	cum	As per annexure 6.1				772
6.22	Boulder / Concrete Block Pitching on Embankment Slopes, In front of Abutment and In Launching Apron complete as per drawings and Technical Specification Section 2500.	cum					-
6.23	Boulder / Concrete Block Pitching in hand woven Wire Crates on Embankment Slopes in front of Abutment and in launching apron complete as per drawings and Technical Specification Section 2500.	cum					-

Annexure 2.1(a) : Template Shape / Component Report (from Mx Roads) : MTM Road														
Station	Cut Volume	Fill Volume	BC Volume	Tack Coat Sur. Area	DBM Volume	Prime Coat Sur. Area	WMM Volume	GSB Volume	Subgrade Volume	<6m Sur. Area	Cut Slope <12 m Sur. Area	> 12 m Sur. Area	RE Wall Sur. Area	R Wall Height
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0														
10	490.3	1.8	3.3	110.0	5.5	110.0	28.6	21.7	65.3		266.0			
15	314.9	15.4	1.6	53.3	2.8		56.0	14.3	10.8		173.0			
20	341.1	17.9	1.6	53.3	2.7	54.0	14.1	10.5	30.9		181.0			
25	344.8	3.0	1.5	50.0	2.6	52.0	13.1	9.8	28.8		173.0			
30	463.8	0.6	1.8	60.0	2.9	58.0	15.1	11.3	33.3	1.0	154.0			
40	1337.8		3.3	110.0	5.5	110.0	28.3	21.1	62.1	33.0		243.0		
50	2449.8		3.3	110.0	5.5	110.0	28.3	21.1	62.1		36.8	54.5		
60	4191.8		3.3	110.0	5.5	110.0	28.3	21.1	62.1			586.0		
70	6283.7		3.3	110.0	5.5	110.0	28.3	21.1	62.1			688.0		
80	7854.5		3.3	110.0	5.5	110.0	28.3	21.1	62.1			665.0		
84	2658.7		1.4	46.7	2.4	48.0	12.4	9.3	27.2		71.0			
90	2108.9		1.8	60.0	3.1	62.0	15.9	11.9	35.0		91.0			
100	3561.2		3.3	110.0	5.5	110.0	28.3	21.1	62.1		165.0			
110	4758.7		3.3	110.0	5.5	110.0	28.3	21.1	62.1		152.0	256.0		
120	5249.7		3.3	110.0	5.5	110.0	28.3	21.1	62.1	33.0		499.0		
130	3755.9		3.3	110.0	5.5	110.0	28.3	21.1	62.1	51.0		437.0		
140	2879.1		3.3	110.0	5.5	110.0	28.3	21.1	62.1	36.0		550.0		
150	2495.8		3.3	110.0	5.5	110.0	28.3	21.1	62.1	44.0		631.0		
160	2061.0		3.3	110.0	5.5	110.0	28.3	21.1	62.1	53.0	99.0			
170	1623.6		3.3	110.0	5.5	110.0	28.3	21.1	62.1		270.0			
180	1156.5	2.5	3.4	113.3	5.7	114.0	29.3	22.0	65.4		332.0			
190	640.0	8.3	3.5	116.7	5.8	116.0	30.3	22.9	68.6		213.0			
200	278.7	5.7	3.5	116.7	5.8	116.0	30.3	22.9	68.6		130.0			
210	123.1		3.5	116.7	5.8	116.0	30.3	22.9	68.6	3.0				
220	107.9		3.5	116.7	5.8	116.0	30.3	22.9	68.6	17.0				
230	161.5	0.5	3.5	116.7	5.8	116.0	30.3	22.9	68.6	44.0				
239	288.0	2.8	3.0	100.0	5.0	100.0	26.2	19.8	59.4		135.0			
240	64.4	0.7	0.5	16.7	0.8	16.0	4.1	3.1	9.3		42.0			
250	444.3	8.5	3.5	116.7	5.8	116.0	30.3	22.9	68.6		212.0			
260	395.9	14.8	3.5	116.7	5.8	116.0	30.3	22.9	68.6		106.0			
270	405.3	9.2	3.5	116.7	5.8	116.0	30.3	22.9	68.6		134.0			
280	357.6	0.2	3.5	116.7	5.8	116.0	30.3	22.9	68.6	28.0				
290	283.1	1.8	3.5	116.7	5.8	116.0	30.3	22.9	68.6	55.0				
300	283.8	6.2	3.5	116.7	5.8	116.0	30.3	22.9	68.6	53.0				
310	327.4	5.5	3.5	116.7	5.8	116.0	30.3	22.9	68.6		73.0			
315	156.4	1.4	1.6	53.3	2.7	54.0	14.0	10.6	31.6	15.0				
315	3.1				0.1	2.0	0.3	0.2	0.7	1.0				
320	157.0	2.9	1.8	60.0	3.1	62.0	16.0	12.1	36.3	34.0				
330	248.3	5.6	3.2	106.7	5.4	108.0	27.9	20.8	56.1	62.0				0.8
330	9.2	0.2	0.1	3.3	0.2	4.0	1.1	0.8	2.3	3.0				0.8
340	195.3	5.6	3.3	110.0	5.5	110.0	27.8	20.4	58.4	60.0				1.3
350	128.3	11.9	3.3	110.0	5.5	110.0	27.8	20.4	58.4	47.0				2.5
360	133.7	15.0	3.3	110.0	5.5	110.0	27.8	20.4	58.4	51.0				2.2
370	186.4	12.2	3.4	113.3	5.7	114.0	29.0	21.7	63.5	56.0				
380	204.4	5.8	3.5	116.7	5.8	116.0	30.3	22.9	68.6	43.0				
389	196.6	5.0	3.1	103.3	5.2	104.0	27.0	20.3	55.6	31.0				
390	17.3	0.8	0.3	10.0	0.4	8.0	2.2	1.6	4.8	3.0				
400	250.1	6.1	3.3	110.0	5.5	110.0	28.3	21.1	62.1	34.0				
410	303.2	1.1	3.3	110.0	5.5	110.0	28.3	21.1	62.1	38.0				
420	319.7	0.3	3.3	110.0	5.5	110.0	28.3	21.1	62.1	41.0				
430	303.7	0.4	3.3	110.0	5.5	110.0	28.3	21.1	62.1	44.0				
440	351.6	0.8	3.3	110.0	5.5	110.0	28.3	21.1	62.1	51.0				
450	311.0	2.7	3.3	110.0	5.5	110.0	28.3	21.1	62.1	44.0				
460	203.5	2.1	3.4	113.3	5.7	114.0	29.3	22.0	65.4	25.0				
465	84.9	0.8	1.8	60.0	3.0	60.0	15.4	11.5	31.0	8.0				0.5
470	52.5	3.5	1.5	50.0	2.4	48.0	12.3	9.1	26.0	5.0				1.9
480	87.2	12.7	3.1	103.3	5.1	102.0	25.9	19.1	54.7	6.0				2.1
486	42.4	7.0	1.8	60.0	2.9	58.0	14.9	11.0	31.5	2.0				1.9
490	28.7	4.4	1.3	43.3	2.1	42.0	10.6	7.8	22.4	3.0				1.8
491	8.0	1.2	0.4	13.3	0.6	12.0	3.0	2.2	6.4	1.0				1.8
500	59.1	7.9	2.7	90.0	4.5	90.0	22.8	16.7	48.0	12.0				1.6
506	43.3	4.9	1.9	63.3	3.2	64.0	16.4	12.0	34.5	12.0				1.5
510	32.3	2.4	1.2	40.0	2.1	42.0	10.4	7.6	21.9	10.0				1.0
510	1.1	0.1			0.1	2.0	0.3	0.2	0.6					1.0
520	175.3	4.8	3.3	110.0	5.6	112.0	28.6	21.4	63.0	48.0				
530	355.4	2.5	3.5	116.7	5.8	116.0	30.4	23.1	69.4		95.0			
540	645.8	2.6	3.4	113.3	5.7	114.0	29.3	22.1	60.1		282.0			
550	1046.5	3.9	3.3	110.0	5.5	110.0	28.3	21.1	62.1			255.0		
560	1348.3	2.3	3.3	110.0	5.5	110.0	28.3	21.1	62.1			524.0		
570	1337.8	2.7	3.3	110.0	5.5	110.0	28.3	21.1	62.1			530.0		
576	691.2	4.4	2.1	70.0	3.5	70.0	18.2	13.6	39.9		67.0			
580	285.0	1.8	1.2	40.0	2.0	40.0	10.3	7.7	23.1		102.0			
580	26.3	0.1	3.3	0.2	4.0		1.1	0.8	2.5		13.0			
590	491.9	0.3	3.2	106.7	5.3	106.0	27.8	21.1	63.6		259.0			
595	139.2	2.1	1.7	56.7	2.9	58.0	15.1	11.5	34.5	17.0				
600	82.0	4.1	1.5	50.0	2.5	50.0	13.0	9.9	29.7	26.0				
610	148.8	5.2	3.1	103.3	5.2	104.0	27.0	20.4	55.8	34.0				
618	72.6	14.7	2.4	80.0	4.0	80.0	20.8	15.6	46.1					
618														
620	6.9	7.8	0.6	20.0	1.0	20.0	5.2	3.9	11.5					
630	30.3	42.8	3.0	100.0	5.1	102.0	26.0	19.5	57.6					
640	39.0	43.6	3.0	100.0	5.1	102.0	26.0	19.5	57.6	3.0				
640	1.3	1.1	0.1	3.3	0.1	2.0	0.7	0.5	1.5					
650	69.4	39.1	3.0	100.0	5.0	100.0	25.6	19.2	56.7	12.0				
655	63.4	10.0	1.7	56.7	2.9	58.0	14.8	11.1	33.2	18.0				
660	80.7	0.3	1.5	50.0	2.6	52.0	13.4	10.2	30.5	25.0				
660	4.9	0.1	3.3	0.1	2.0		0.7	0.5	1.6	1.0				
670	274.6	4.4	3.5	116.7	5.8	116.0	30.2	22.9	68.4		82.0			
680	424.4	6.3	3.4	113.3	5.7	114.0	29.0	21.6	58.4		255.0			
690	520.2	4.6	3.3	110.0	5.5	110.0	27.8	20.4	58.4		343.0		0.8	
700	556.8	2.3	3.4	113.3	5.7	114.0	29.0	21.7	63.5		341.0		0.8	
710	620.1	0.1	3.5	116.7	5.8	116.0	30.2	22.9	68.6		348.0			
720	503.0		3.5	116.7	5.8	116.0	30.3	23.0	69.0		230.0			
730	218.3	2.5	3.5	116.7	5.8	116.0	30.4	23.1	69.4					
740	44.6	60.2	3.4	113.3	5.7	114.0	29.0	21.7	58.2					
750	25.9	64.3	3.4	113.3	5.7	114.0	29.0	21.7	63.8				2.2	
760	82.3	7.7	3.5	116.7	5.8	116.0	30.3	23.1	69.4					
770	117.0	1.5	3.5	116.7	5.8	116.0	30.3	23.1	69.4	1.0				
780	136.9	1.2	3.5	116.7	5.8	116.0	30.3	23.1	69.4	10.0				
781	14.8	0.1	0.3	10.0	0.6	12.0	3.0	2.2	6.8	2.0				
790	105.2	3.0	3.0	100.0</										

Sl. No.	Proposed Chainage (km)	Designation	Pipe Length	Excavation	Back Filling	Conc. Foundation	Conc. Substructure	Levelling coarse	Filter Media	Painting	Lettering
24	29.810	1*1.2	12.50	162.72	179.18	14.58	24.00	3.88	33.84	2.28	50
25	48.260	1*1.2	12.50	164.31	179.18	14.58	24.00	3.88	33.84	2.28	50
26	54.750	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
27	54.530	1*1.2	12.50	165.49	179.18	14.58	24.00	3.88	33.84	2.28	50
28	56.120	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
29	56.060	1*1.2	12.50	138.89	137.21	12.96	20.40	3.47	30.24	2.28	50
30	29.350	1*1.2	12.50	110.56	137.21	12.96	20.40	3.47	30.24	2.28	50
31	29.903	1*1.2	12.50	125.13	137.21	12.96	20.40	3.47	30.24	2.28	50
32	26.900	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
33	77.710	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
34	4.232	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
35	78.090	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
36	56.770	1*1.2	12.50	132.63	137.21	12.96	20.40	3.47	30.24	2.28	50
37	77.200	1*1.2	12.50	132.48	137.21	12.96	20.40	3.47	30.24	2.28	50
38	79.270	1*1.2	12.50	124.49	137.21	12.96	20.40	3.47	30.24	2.28	50
39	48.650	1*1.2	12.50	131.92	137.21	12.96	20.40	3.47	30.24	2.28	50
40	8.500	1*1.2	12.50	123.07	137.21	12.96	20.40	3.47	30.24	2.28	50
41	72.855	1*1.2	12.50	115.50	137.21	12.96	20.40	3.47	30.24	2.28	50
42	22.660	1*1.2	12.50	119.36	137.21	12.96	20.40	3.47	30.24	2.28	50
43	68.755	1*1.2	12.50	121.27	137.21	12.96	20.40	3.47	30.24	2.28	50
44	10.200	1*1.2	12.50	130.57	137.21	12.96	20.40	3.47	30.24	2.28	50
45	50.980	1*1.2	12.50	129.90	137.21	12.96	20.40	3.47	30.24	2.28	50
46	57.740	1*1.2	12.50	128.64	137.21	12.96	20.40	3.47	30.24	2.28	50
47	1.500	1*1.2	12.50	132.21	137.21	12.96	20.40	3.47	30.24	2.28	50
48	58.775	1*1.2	12.50	126.99	137.21	12.96	20.40	3.47	30.24	2.28	50

Sl. No.	Proposed Chainage	Designation	Excavation	Rcc for Box	Approach slab rcc	Exp. Gap	Lettering	Drainage Spout	Painting	Levelling course	Excavation for Return Wall	RCC for Return Wall	Filter Media	Backfill	RCC for Counterfort	RCC for Wall
	(km)		m ³	m ³	m ³	lm	nos.	nos.	m ²	m ³	m ³	m ³	m ³	m ³	m ³	m ³
28	65.425	1.22.00	71.222	57.95	27.44	22.40	50.00	4.00	9.88	16.30	59.77	22.24	20.16	91.48	0.00	0.00
29	30.687	1.22.00	59.703	52.19	24.50	20.00	50.00	4.00	9.88	14.59	57.64	22.24	20.16	91.48	0.00	0.00
30	25.000	1.22.00	86.800	59.39	28.18	23.00	50.00	4.00	9.88	16.73	66.27	22.24	20.16	91.48	0.00	0.00
31	70.290	1.22.00	69.702	55.07	25.97	21.20	50.00	4.00	9.88	15.44	60.95	22.24	20.16	91.48	0.00	0.00
32	73.220	1.22.00	58.452	56.51	26.71	21.80	50.00	4.00	9.88	15.87	54.31	22.24	20.16	91.48	0.00	0.00
33	52.025	1.22.00	49.103	53.73	25.28	20.64	50.00	4.00	9.88	15.04	51.00	22.24	20.16	91.48	0.00	0.00
34	47.760	1.22.00	74.605	53.85	25.35	20.69	50.00	4.00	9.88	15.08	64.43	22.24	20.16	91.48	0.00	0.00
35	66.713	1.22.00	47.739	52.19	24.50	20.00	50.00	4.00	9.88	14.59	51.09	22.24	20.16	91.48	0.00	0.00
36	75.960	1.22.00	60.760	52.19	24.50	20.00	50.00	4.00	9.88	14.59	58.22	22.24	20.16	91.48	0.00	0.00
37	76.290	1.22.00	77.482	59.39	28.18	23.00	50.00	4.00	9.88	16.73	61.84	22.24	20.16	91.48	0.00	0.00
38	78.660	1.22.00	46.081	59.39	28.18	23.00	50.00	4.00	9.88	16.73	46.89	22.24	20.16	91.48	0.00	0.00
39	61.160	1.22.00	66.472	52.19	24.50	20.00	50.00	4.00	9.88	14.59	61.34	22.24	20.16	91.48	0.00	0.00
40	77.410	1.22.00	40.986	53.15	24.99	20.40	50.00	4.00	9.88	14.87	46.95	22.24	20.16	91.48	0.00	0.00
41	71.790	1.22.00	48.013	52.19	24.50	20.00	50.00	4.00	9.88	14.59	51.24	22.24	20.16	91.48	0.00	0.00

SL. No.	DOWNSTREAM PROTECTION WORKS										UPSTREAM PROTECTION WORKS									
	Proposed Chainage		Excavation		PCC M15		RCC for flooring+ wall +block		Rcc in Front and Back Curtain Wall		RCC Total		Hand woven Wire Crates		Reinforce ment		Excavation		Excavation for Catchpit	
	(km)	m ³	m ³	m ³	m ³	m ³	m ³	m ³	m ³	m ³	m ³	m ³	m ³	m ³	MT	MT	m ³	m ³	m ³	m ³
74	14.045	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
75	1.005	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
76	56.170	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
77	1.190	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
78	68.205	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
79	57.260	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
80	57.590	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
81	0.465	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
82	0.590	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
83	77.640	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
84	13.955	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
85	4.665	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.29	75.80	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
86	10.360	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
87	29.080	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
88	46.570	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
89	83.635	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
90	54.290	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
91	51.055	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
92	4.465	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
93	34.140	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
94	36.055	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
95	14.750	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
96	52.325	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
97	76.560	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
98	19.530	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
99	36.020	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
100	48.205	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
101	49.210	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
102	4.985	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
103	51.980	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
104	85.480	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
105	69.030	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
106	45.953	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
107	20.030	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
108	19.123	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
109	51.880	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20
110	55.430	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	1.20

Sl. No.	DOWNSTREAM PROTECTION WORKS										UPSTREAM PROTECTION WORKS									
	Excavation	PCC M15	RCC for flooring+ wall +block	Rcc in Front and Back Curtain Wall	RCC Total	Hand woven Wire Crates	Reinforce ment	Excavation	Excavation for Catchpitt	Excavation Total	PCC M15	RCC	RCC For Catchpitt	RCC Total	Hand woven Wire Crates	Reinforce ment	Reinforce ment for Catchpitt	Reinforce ment Total		
	(km)	m ³	m ³	m ³	m ³	m ³	MT	m ³	m ³	m ³	m ³	m ³	m ³	m ³	m ³	MT	MT	MT		
222	52.515	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
223	11.800	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	45.40	95.92	5.03	19.03	5.57	24.59	13.80	0.99	0.45	1.44	
224	30.360	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	34.88	85.40	5.03	19.03	4.02	23.04	13.80	0.99	0.32	1.32	
225	40.580	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
226	15.645	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
227	53.415	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
228	55.120	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
229	15.750	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
230	79.875	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
231	1.783	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
232	76.740	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
233	51.240	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
234	45.190	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
235	55.755	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
236	15.090	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
237	6.272	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	26.59	77.10	5.03	19.03	2.80	21.82	13.80	0.99	0.22	1.22	
238	24.905	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	25.25	75.76	5.03	19.03	2.60	21.63	13.80	0.99	0.21	1.20	
239	79.680	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	86.39	136.90	5.03	19.03	11.60	30.63	13.80	0.99	0.93	1.92	
240	40.240	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52	75.60	126.12	5.03	19.03	10.01	29.04	13.80	0.99	0.80	1.80	
241	41.440	43.17	2.58	6.07	4.51	10.58	27.60	0.53	50.52			5.03	19.03		52.33	33.26			292	
	Total	10403	622	1463	1086	2549	6652	127			18407	1212								

Detailed Calculation For Upstream Protection Works for 2mx2m Double cell Box Culvert

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m ²)	H (m) / Thick. (m)	Volume/Qty (m ³)	SUB HEAD TOTAL
	Segment 1									
5.01	Earthwork In excavations:									
	For RCC Rigid flooring (M-30)		m ³	1.00	3.00	6.70	20.10	0.60	12.06	12.06
	For Hand woven wire crates in flexible apron for protection work in upstream		m ³	1.00	3.00	7.70	23.10	1.30	30.03	30.03
	For curtain wall in segment 1		m ³	1.00	7.70	1.70		1.40	18.33	18.33
									Sub Total	60.42
	70% in Ordinary Soil									42.29
	30% in Hard Rock									18.12
5.02	PCC M-15 below curtain wall	100 mm thick.	m ³	1.00	7.70	1.70		0.10	1.31	1.31
5.02	PCC M-15 below RCC flooring	100 mm thick.	m ³	1.00	3.00	6.70	20.10	0.10	2.01	2.01
									Sub Total	3.32
5.04	RCC rigid flooring M-30			1.00	3.00	6.70	20.10	0.20	4.02	4.02
5.04	RCC Side walls over rigid flooring M-30	0.2 m thick.	m ³	2.00	3.16	0.20		1.00	1.26	1.26
									Sub Total	5.28
5.04	Rcc Curtain wall 1		m ³	1.00	7.70		0.84		6.47	6.47
5.20	Hand woven wire crates in flexible apron for protection work	1000 mm thick.	m ³	1.00	3.00	7.70	23.10	1.00	23.10	23.10
5.07	Reinforcement in Rcc curtain wall		t	1.00	6.47	@50kg/m3 of concrete			323.40	0.323
5.07	Reinforcement in RCC flooring		t	1.00	5.28	@50kg/m3 of concrete			264.25	0.264
	Segment-2									
5.01	Earthwork In excavations:									
	Rcc slope protection structure in segment 2		m ³	1.00	9.71	5.70		0.60	33.21	33.21
	For curtain wall in segment 2		m ³	1.00	5.70	0.40		1.50	3.42	3.42
									Sub Total	33.21
	50% in Ordinary Soil									16.60
	50% in Hard rock									16.60
5.02	Pcc below Curtain wall			1.00	5.70	0.40		0.10	0.23	0.23
5.02	Pcc below Concrete for slope protection structure	(per meter)	m ³	1.00	9.71	5.70		0.10	5.53	5.53
										5.76
5.04	Rcc Concrete for slope protection structure	(per meter)	m ³	1.00	9.71	5.70		0.20	11.07	11.07
	RCC Side walls over rigid flooring M-30	(per meter)	m ³	2.00	9.71	0.20		1.00	3.88	3.88
5.04	Rcc Curtain wall		m ³	1.00	5.70	0.3		1.50		2.57
5.04	Buffle pier		m ³	44.00	0.20	0.20		0.20	0.352	0.352
									Sub Total	17.87
5.07	Reinforcement in Rcc curtain wall		t	1.00	2.5650	@50kg/m3 of concrete			128.25	0.128
5.07	Reinforcement in RCC slope protection structure		t	1.00	15.31	@50kg/m3 of concrete			765.27	0.765
	Segment 3									
5.01	Earthwork in excavations:									
	For catchpit		m ³	1.00	2.55	6.30	16.07	1.30	20.88	20.88
	50% in soil									10.44
	50% in ordinary rock									10.44
5.02	PCC M-15 below Catchpit		m ³	1.00	2.45	6.20	15.19	0.10	1.52	1.52
5.04	Rcc M-30 Catchpit walls	A1 (per meter)	m ³	2.00	2.00	0.30		1.00	1.20	1.20
		A2 (per meter)	m ³	1.00	5.90	0.30		1.00	1.77	1.77
									Sub Total	2.97
5.04	RCC M-30 Catchpit base		m ³	1.00	2.45	3.90		0.20	1.91	1.91
5.07	Reinforcement in Rcc catchpit walls		t	1.00	2.97	@50kg/m3 of concrete			148.50	0.149
5.07	Reinforcement in Rcc catchpit base		t	1.00	1.91	@50kg/m3 of concrete			95.55	0.096

Detailed Calculation For Downstream Protection Works for 3mx3m Double cell Box Culvert

ITEM NO.	ITEM	DESCRIPTION	UNIT	NO.	L (m)	B (m)	AREA (m ²)	H (m) / Thick. (m)	Volume/Q ty (m ³)	SUB HEAD TOTAL
5.1	Earthwork in excavations for protection works									
	For RCC Rigid flooring (M-30)		m ³	1.00	5.00	8.70	43.50	0.30	13.05	13.05
	For Hand woven wire crates in flexible apron for protection work		m ³	1.00	6.00	9.70	58.20	1.00	58.20	58.20
	For front curtain wall		m ³	1.00	9.70	1.70		1.30	21.44	21.44
	For back curtain wall		m ³	0.00	7.70	0.30		1.00	0.00	0.00
										92.69
	70% in soil									64.88
	30% In ordinary rock									27.81
Outlet/downstream Protection Work										
5.04	RCC channel/flooring M-30	0.2m thick	m ³	1.00	5.00	8.70		0.20	8.70	8.70
5.04	RCC Side wall of Rigid Flooring M-30			2.00	5.10	0.20		1.00	2.04	2.04
5.04	RCC concrete block		m ³	60.00	0.30	0.20		0.30	1.08	1.08
		Sub Total								11.82
5.04	RCC in Front Curtain wall		m ³	1.00	9.70		0.81		7.86	7.86
5.04	RCC in Back Curtain wall		m ³	1.00	7.70	0.30		1.00	2.31	2.31
		Sub Total								10.17
5.02	PCC M-15 below RCC Rigid flooring	0.1m thick	m ³	1.00	5.00	8.70		0.10	4.35	4.35
5.02	PCC M-15 below front Curtain wall		m ³	1.00	9.70	1.70		0.10	1.65	1.65
5.20	Hand woven wire crates in flexible apron for protection work	1m thick	m ³	1.00	6.00	9.70		1.00	58.20	58.20
5.07	Reinforcement in RCC flooring		Mt	1.00	11.82	.@50kg/m ³ of concrete			590.98	0.591
5.07	Reinforcement in curtain wall		Mt	1.00	10.17	.@50kg/m ³ of concrete			508.35	0.508

Summary of Quantities for bridges of MTM Road

Item No. as per Quantity Estimates	Item No. as per BOQ	Description	Unit	Steel Composite Girder at KM 31.565	RCC voided Slab at KM. 34.485	Composite Girder at KM. 38.722	Composite Girder at KM. 41.500	RCC Voided Slab at KM 79.045	RCC Slab at KM 86.795	Total quantity	Item No. in Rate Analysis	Remarks
1	2	3	4	5	6	7	8	9	10	11	12	13
	iii	Metallic Guide Bearings, having Capacity of								0		
		25 T ± 12.5 T	nos							0	As per Market	
		50 T ± 12.5 T	nos							0	As per Market	
		75 T ± 12.5 T	nos							0	As per Market	
		100 T ± 12.5 T	nos		1			1		2	As per Market	
6.11	6.12	Expansion Joints, of following Type as per drawings and Technical Specification Section 2600								0		
	a	Filler	m						27	27	14.18 (ii)	20mm thick premoulded filler joint
	b	Asphaltic Plug	m							0	14.19	
	c	Strip Seal Type capable of taking								0		
	i	40 mm movement	m	12	25	24	12	26		100	14.22 A New	
	ii	80 mm movement	m	24		24				48	14.23	
	iii	160 mm movement	m				60			60	14.24	
6.12	6.13	Helical Void Former for Voided Concrete Slabs including 20 mm diameter tie rod and sealing joints as per drawings and Technical Specification Section 2300 and Additional Technical Specification A - 4.	m		84			108		192	14.14	
6.13	6.14	Drainage Spouts as per Technical Specification Section 2700	nos	45	17	24	130	20	13	250	14.9	
6.14	6.15	Approach Slab (RCC, M 30 Grade) including reinforcement as per drawings and Technical Specification Section 2700	cum	29	33	29	29	32	33	187	14.11	
6.15	6.16	RCC (M 30 Grade) comprising cast in-situ Post and Precast Hand rails including reinforcement complete as per drawings and Technical Specification Section 1600, 1700 and 2700.	Rm	106	42	60	322	53	34	614	14.6	
6.16	6.17	RCC Crash Barrier (M 40 Grade) excluding cost of reinforcement complete as per drawings and Technical Specification Section 1700 and 2700	cum	29	15	16	88	13	8	173	8.22 B	
6.17	6.18	Filling of Selected Imported granular materials below structures as per drawings and Technical Specification Clause 305, 1000 and 2900	Cum							0	13.9 A	Granular material
6.18	6.19	Weep holes (AC/PVC/HDPE) in Earth Retaining Structures (Abutments, Spayed Wing wall, Median Wall, Retaining Wall etc.) complete as per drawings and Technical Specification Section 2700.	nos	247	388	204	386	422	295	1943	13.8	
6.19	6.20	Back filling behind Earth Retaining Structures (abutments, spayed wing walls, retaining wall, box cell structures etc.) Complete as per drawings and Technical Specification Clause 305.	cum	1169	1201	608	215	2518	2176	7887	13.9 A	
6.20	6.21	Filter Media behind Earth Retaining Structures (abutment, spayed wing wall, retaining wall, return wall etc.) complete as per Technical Specification Section 2500	cum	112	147	71	40	199	203	772	13.10	
6.21	6.29	Boulder Grouted with Cement Mortar (1 : 3) in annular space around footings complete as per drawings and Technical Specification 304 and 2100	cum	314	26	54	3076	84	81	3635	12.43	
6.22	6.30	PCC M-15 in annular space around footings complete as per drawings and Technical Specification 304, 1700 and 2100.	cum	87	39	31	141	6	29	332	12.8 A	
6.23	6.32	Preparation of rock foundation surface and filling/ sealing of seams with cement grout or mortar complete as per drawings and Technical Specifications Sections 304 and 2805.	sqm	309	230	205	1388	280	243	2655	16.4+16.5(b)+(16.1)²	
6.24	6.33	Letting and Number of Bridges as per drawings and Additional Technical Specification A-3	Nos	100	100	100	100	100	40	540	8.3 (ii)	
6.25	6.34	Providing and applying two coats of water based cement paint on Crash barrier and Railing complete as per Technical Specifications Section 800.	sqm	530	420	300	1610	500	340	3700	14.16	
6.26	6.35	Carrying out sub soil investigation / confirmatory boreholes at specified foundation locations before commencement of construction complete as per drawings and Technical Specifications section 2400 or as directed by Engineer.										
	a	In soft rock	Lm	13	10	10	18	10	10	71	As per Market	
	b	Hard rock	Lm	44	32	28	112	30	32	278	As per Market	

QUANTITY CALCULATION FOR BRIDGE AT KM - 38.722

Number of Spans : 1

Longitudinal Girder

169.1

Longitudinal Girder Mid Section:

Top flange :	400 x	20 x	7880	0.063	m ³
Web :	1548 x	12 x	7880	0.146	m ³
Bottom Flange :	600 x	32 x	7880	0.151	m ³

Longitudinal Girder End Section:

Top flange :	400 x	16 x	15120	0.097	m ³
Web :	1559 x	12 x	15120	0.283	m ³
Bottom Flange :	600 x	25 x	15120	0.227	m ³
Total =				0.967	m ³

Weight of single longitudinal Girder	7.592	MTon
Additional weight due to splice plates and stiffeners	1.518	MTon
Number of Longitudinal Girder =	5	Nos
Total weight of Longitudinal Girder =	45.553	MTon

End Cross Girder :

Top flange :	350 x	20 x	2000	0.014	m ³
Web :	1268 x	16 x	2000	0.041	m ³
Bottom Flange :	350 x	20 x	2000	0.014	m ³
Total =				0.069	m ³

Total weight of a Panel of Cross Girder =	0.538	MTon.
Additional weight due to connection =	0.108	MTon.
Total number of Cross Girder Panel	8	Nos
Total weight of Cross Girder =	5.168	MTon

Vertical Bracing:

weight/m of ISA 100x100x8 as horizontal Bracing =	0.012	Mton/m
Total length of horizontal member (ISA: 100x100x8)	110.4	m
Total length of cross member (ISA: 100x100x8)	144.0	m
Total weight of cross bracing	3.078	MTon

Horizontal Bracing:

weight/m of ISA 100x100x8 as horizontal Bracing =	0.012	Mton/m
Total length of angle section on horizontal bracing =	110.88	m
Total weight of horizontal bracing:	1.342	MTon

Bracket for Seismic restrainer:

Weight of seismic restrainer bracket	3.5	MTon
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Total weight of Steel for 23m Composite Girder:	58.640	MTon
	0.222	Mton/m ²

Weight of Shear Connector Studs

Weight of a single Studs	0.0008	MTon
Total Number of Studs per span :	2300	Nos.
Total weight of shear connector Struds	1.773	MTon

Annexure -8.01

Chainage		Length	Type of Cross Section
From	To		
0+000	0+040	40	I
0+040	0+160	120	III
0+160	5+950	5790	I
5+950	5+970	20	II
5+970	6+350	380	I
6+350	6+370	20	II
6+370	7+480	1110	I
7+480	7+500	20	III
7+500	7+560	60	II
7+560	7+810	250	I
7+810	7+850	40	III
7+850	7+860	10	II
7+860	8+050	190	I
8+050	8+075	25	III
8+075	8+130	55	I
8+130	8+155	25	III
8+155	8+690	535	I
8+690	9+000	310	II
9+000	9+195	195	I
9+195	9+270	75	II
9+270	9+485	215	I
9+485	9+490	5	II
9+490	9+630	140	I
9+630	9+640	10	III
9+640	9+700	60	I
9+700	9+715	15	III
9+715	9+800	85	I
9+800	10+040	240	III
10+040	10+100	60	I
10+100	10+190	90	III
10+190	10+960	770	I
10+960	10+990	30	II
10+990	11+095	105	I
11+095	11+105	10	III
11+105	11+220	115	I
11+220	11+380	160	II
11+380	11+560	180	I
11+560	11+610	50	III
11+610	11+790	180	I
11+790	11+850	60	II
11+850	11+980	130	I
11+980	12+255	275	II
12+255	13+250	995	I
13+250	13+320	70	III
13+320	15+980	2660	I
15+980	16+030	50	II
16+030	16+680	650	I
16+680	16+730	50	II
16+730	17+000	270	I
17+000	17+230	230	II
17+230	18+350	1120	I
18+350	18+360	10	III
18+360	18+680	320	I
18+680	18+690	10	III
18+690	19+470	780	I
19+470	20+215	745	I
20+215	20+220	5	III

Summary of Gabion Wall Data			
Cl. Ht (m)	X Area (m ²)	Excavation (m ³)	Backfill (m ³)
0.0	0	0	0
0.5	1.50	0.57	0.29
1.0	2.75	2.79	1.15
1.5	4.13	6.38	2.60
2.0	4.625	7.65	3.54
2.5	9.25	12.36	4.62
3.0	9.75	14.00	5.85
3.5	12.75	17.82	7.22
4.0	13.25	19.58	9.48
4.5	16.75	24.02	11.64
5.0	17.25	25.90	14.95
5.5	21.25	30.96	17.65
6.0	21.75	32.95	21.49
6.5	26.25	38.63	24.73
7.0	26.75	40.75	29.11
7.5	32.25	51.30	36.13
8.0	32.75	53.78	41.30

Station/ Chainage (m)	Type of Retaining Structure	Retain Ht.	Retain Length	Culvert Position	X-Area of GW (m²)	Surface Area of RE Wall (m²)	Volume of GW (m³)	RE Wall Actually Provided GW (m²)	Addl Qty GW in place replaced RE Wall (m³)	Deducted Qty GW at Culvert location (m³)	Esca. Qty for GW (m³)	Backfill Qty for GW (m³)	Esca. Qty for REW (m³)	Backfill Qty for REW (m³)
2670	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2680	Para W	0.0	0.00		0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
2690	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2700	GW	2.0	10.00		4.63	0.00	46.25				76.54	35.36	0.00	0.00
2710	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2720	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2730	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2730.255	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2735.387	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2740	Para W	0.0	0.00		0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.00
2750	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2750.387	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2760	GW	1.5	9.61		4.13	0.00	39.70				61.38	24.98	0.00	0.00
2770	GW	5.0	10.00		17.25	0.00	172.50			138.00	258.96	149.47	0.00	0.00
2780	GW	3.0	10.00		9.75	0.00	97.50				139.97	58.46	0.00	0.00
2790	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
2797.308	GW	1.5	7.31		4.13	0.00	30.18				46.66	18.99	0.00	0.00
2800	GW	1.3	7.88		4.13	0.00	11.12				17.19	8.85	0.00	0.00
2810	GW	1.0	10.00		2.75	0.00	27.50				27.95	11.55	0.00	0.00
2820	GW	1.0	10.00		2.75	0.00	27.50				27.95	11.55	0.00	0.00
2830	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00
2840	GW	3.0	10.00		9.75	0.00	97.50			78.00	139.97	58.46	0.00	0.00
2844.65	GW	3.5	4.65		12.75	0.00	59.28				82.86	33.56	0.00	0.00
2850	GW	3.0	5.35		9.75	0.00	52.16				74.88	31.27	0.00	0.00
2859.65	GW	2.5	9.65		9.25	0.00	89.26				119.26	44.57	0.00	0.00
2860	GW	2.5	0.35		9.25	0.00	3.24				4.33	1.62	0.00	0.00
2864.15	GW	2.5	4.15		9.25	0.00	38.39				51.29	19.17	0.00	0.00
2870	GW	2.0	5.85		4.63	0.00	27.06				44.78	20.69	0.00	0.00
2880	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00
2890	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2896.197	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2900	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2900.692	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2910	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2915.692	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2920	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2929.243	GW	1.0	9.24		2.75	0.00	25.42				25.83	10.67	0.00	0.00
2930	GW	1.0	0.76		2.75	0.00	2.08				2.12	0.87	0.00	0.00
2936.788	GW	1.0	6.29		2.75	0.00	18.67				18.97	7.84	0.00	0.00
2940	GW	1.0	3.21		2.75	0.00	8.83				8.98	3.71	0.00	0.00
2950	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2951.788	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2960	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2965.288	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2970	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
2980	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
2990	GW	4.0	10.00		13.25	0.00	132.50			106.00	195.77	94.82	0.00	0.00
3000	GW	5.0	10.00		17.25	0.00	172.50				258.96	149.47	0.00	0.00
3010	GW	4.5	10.00		16.75	0.00	167.50				240.20	116.42	0.00	0.00
3013.695	GW	4.5	3.70		16.75	0.00	61.89				88.75	43.02	0.00	0.00
3013.795	GW	5.0	0.10		17.25	0.00	1.72				2.59	1.49	0.00	0.00
3013.796	GW	5.0	0.00		17.25	0.00	0.00				0.00	0.00	0.00	0.00
3020	GW	5.0	6.20		17.25	0.00	107.02				160.66	92.73	0.00	0.00
3028.795	GW	4.5	8.80		16.75	0.00	147.32				211.25	102.39	0.00	0.00
3030	GW	4.5	1.20		16.75	0.00	20.18				28.94	14.03	0.00	0.00
3038.438	GW	3.5	8.44		4.13	0.00	34.85				53.87	21.92	0.00	0.00
3040	GW	3.5	1.56		4.13	0.00	6.45				9.57	4.06	0.00	0.00
3050	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3060	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3070	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3080	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3090	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3096.873	GW	1.5	6.87		4.13	0.00	28.39				43.88	17.85	0.00	0.00
3100	GW	1.5	3.13		4.13	0.00	12.91				19.96	8.12	0.00	0.00
3106.516	GW	2.0	6.52		4.63	0.00	30.14				49.87	23.04	0.00	0.00
3110	GW	3.0	3.48		9.75	0.00	33.97				48.76	20.37	0.00	0.00
3120	GW	4.5	10.00		16.75	0.00	167.50			134.00	240.20	116.42	0.00	0.00
3121.516	GW	4.5	1.52		16.75	0.00	25.39			134.00	36.41	17.65	0.00	0.00
3130	GW	3.0	8.48		9.75	0.00	87.72				118.75	49.59	0.00	0.00
3140	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3146.377	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3146.477	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3146.478	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3150	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3160	GW	1.0	10.00		2.75	0.00	27.50				27.95	11.55	0.00	0.00
3161.477	GW	1.0	1.48		2.75	0.00	4.06				4.33	1.71	0.00	0.00
3168.977	GW	1.5	7.50		4.13	0.00	30.98				47.89	19.49	0.00	0.00
3170	GW	1.5	1.02		4.13	0.00	4.72				6.53	2.66	0.00	0.00
3180	GW	1.0	10.00		2.75	0.00	27.50				27.95	11.55	0.00	0.00
3190	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
3200	GW	3.5	10.00		12.75	0.00	127.50				178.20	72.17	0.00	0.00
3205.206	GW	3.5	5.21		12.75	0.00	66.38				92.77	37.57	0.00	0.00
3210	GW	3.0	4.79		9.75	0.00	46.74				67.30	28.02	0.00	0.00
3220	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00
3230	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3237.348	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3240	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3250	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3260	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00
3268.008	GW	2.5	8.01		9.25	0.00	74.07				98.96	36.99	0.00	0.00
3270	GW	2.5	1.99		9.25	0.00	18.41				24.62	9.20	0.00	0.00
3280	GW	3.5	10.00		12.75	0.00	127.50				178.20	72.17	0.00	0.00
3290	GW	2.0	10.00		4.63	0.00	46.25				76.54	35.36	0.00	0.00
3300	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3303.612	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
3310	GW	1.0	6.39		2.75	0.00	17.57				17.85	7.38	0.00	0.00
3320	GW	2.0	10.00		4.63	0.00	46.25				76.54	35.36	0.00	0.00
3330	GW	3.0	10.00		9.75	0.00	97.50				139.97	58.46	0.00	0.00
3338.389	GW	4.0	8.39		13.25	0.00	111.15				164.23	79.55	0.00	0.00
3340	GW	4.0	1.61		13.25	0.00	21.35				31.54			

Station/ Chainage (m)	Type of Retaining Structure	Retain Ht.	Retain Length	Culvert Position	X-Area of GW (m ²)	Surface Area of RE Wall (m ²)	Volume of GW (m ³)	RE Wall Actually Provided GW (m ²)	Add Qty GW in place replaced RE Wall (m ³)	Deducted Qty GW at Culvert location (m ³)	Emas. Qty for GW (m ³)	Backfill Qty for GW (m ³)	Emas. Qty for REW (m ³)	Backfill Qty for REW (m ³)
9140	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9150	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9160	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9170	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9180	Para W	0.0	0.00		0.00	0.00	0.00				0.00	0.00	0.00	0.00
9190	GW	2.5	10.00		9.25	0.00	92.50				0.00	0.00	0.00	0.00
9200	GW	3.0	10.00		9.75	0.00	97.50				0.00	0.00	0.00	0.00
9210	GW	3.5	10.00		10.25	0.00	102.50				0.00	0.00	0.00	0.00
9220	GW	4.0	10.00		10.75	0.00	107.50				0.00	0.00	0.00	0.00
9230	GW	4.5	10.00		11.25	0.00	112.50				0.00	0.00	0.00	0.00
9240	GW	5.0	10.00		11.75	0.00	117.50				0.00	0.00	0.00	0.00
9250	GW	5.5	10.00		12.25	0.00	122.50				0.00	0.00	0.00	0.00
9260	GW	6.0	10.00		12.75	0.00	127.50				0.00	0.00	0.00	0.00
9270	GW	6.5	10.00		13.25	0.00	132.50				0.00	0.00	0.00	0.00
9280	GW	7.0	10.00		13.75	0.00	137.50				0.00	0.00	0.00	0.00
9290	GW	7.5	10.00		14.25	0.00	142.50				0.00	0.00	0.00	0.00
9300	GW	8.0	10.00		14.75	0.00	147.50				0.00	0.00	0.00	0.00
9310	GW	8.5	10.00		15.25	0.00	152.50				0.00	0.00	0.00	0.00
9320	GW	9.0	10.00		15.75	0.00	157.50				0.00	0.00	0.00	0.00
9330	GW	9.5	10.00		16.25	0.00	162.50				0.00	0.00	0.00	0.00
9340	GW	10.0	10.00		16.75	0.00	167.50				0.00	0.00	0.00	0.00
9350	GW	10.5	10.00		17.25	0.00	172.50				0.00	0.00	0.00	0.00
9360	GW	11.0	10.00		17.75	0.00	177.50				0.00	0.00	0.00	0.00
9370	GW	11.5	10.00		18.25	0.00	182.50				0.00	0.00	0.00	0.00
9380	GW	12.0	10.00		18.75	0.00	187.50				0.00	0.00	0.00	0.00
9390	GW	12.5	10.00		19.25	0.00	192.50				0.00	0.00	0.00	0.00
9400	GW	13.0	10.00		19.75	0.00	197.50				0.00	0.00	0.00	0.00
9410	GW	13.5	10.00		20.25	0.00	202.50				0.00	0.00	0.00	0.00
9420	GW	14.0	10.00		20.75	0.00	207.50				0.00	0.00	0.00	0.00
9430	GW	14.5	10.00		21.25	0.00	212.50				0.00	0.00	0.00	0.00
9440	GW	15.0	10.00		21.75	0.00	217.50				0.00	0.00	0.00	0.00
9450	GW	15.5	10.00		22.25	0.00	222.50				0.00	0.00	0.00	0.00
9460	GW	16.0	10.00		22.75	0.00	227.50				0.00	0.00	0.00	0.00
9470	GW	16.5	10.00		23.25	0.00	232.50				0.00	0.00	0.00	0.00
9480	GW	17.0	10.00		23.75	0.00	237.50				0.00	0.00	0.00	0.00
9490	GW	17.5	10.00		24.25	0.00	242.50				0.00	0.00	0.00	0.00
9500	GW	18.0	10.00		24.75	0.00	247.50				0.00	0.00	0.00	0.00
9510	GW	18.5	10.00		25.25	0.00	252.50				0.00	0.00	0.00	0.00
9520	GW	19.0	10.00		25.75	0.00	257.50				0.00	0.00	0.00	0.00
9530	GW	19.5	10.00		26.25	0.00	262.50				0.00	0.00	0.00	0.00
9540	GW	20.0	10.00		26.75	0.00	267.50				0.00	0.00	0.00	0.00
9550	GW	20.5	10.00		27.25	0.00	272.50				0.00	0.00	0.00	0.00
9560	GW	21.0	10.00		27.75	0.00	277.50				0.00	0.00	0.00	0.00
9570	GW	21.5	10.00		28.25	0.00	282.50				0.00	0.00	0.00	0.00
9580	GW	22.0	10.00		28.75	0.00	287.50				0.00	0.00	0.00	0.00
9590	GW	22.5	10.00		29.25	0.00	292.50				0.00	0.00	0.00	0.00
9600	GW	23.0	10.00		29.75	0.00	297.50				0.00	0.00	0.00	0.00
9610	GW	23.5	10.00		30.25	0.00	302.50				0.00	0.00	0.00	0.00
9620	GW	24.0	10.00		30.75	0.00	307.50				0.00	0.00	0.00	0.00
9630	GW	24.5	10.00		31.25	0.00	312.50				0.00	0.00	0.00	0.00
9640	GW	25.0	10.00		31.75	0.00	317.50				0.00	0.00	0.00	0.00
9650	GW	25.5	10.00		32.25	0.00	322.50				0.00	0.00	0.00	0.00
9660	GW	26.0	10.00		32.75	0.00	327.50				0.00	0.00	0.00	0.00
9670	GW	26.5	10.00		33.25	0.00	332.50				0.00	0.00	0.00	0.00
9680	GW	27.0	10.00		33.75	0.00	337.50				0.00	0.00	0.00	0.00
9690	GW	27.5	10.00		34.25	0.00	342.50				0.00	0.00	0.00	0.00
9700	GW	28.0	10.00		34.75	0.00	347.50				0.00	0.00	0.00	0.00
9710	GW	28.5	10.00		35.25	0.00	352.50				0.00	0.00	0.00	0.00
9720	GW	29.0	10.00		35.75	0.00	357.50				0.00	0.00	0.00	0.00
9730	GW	29.5	10.00		36.25	0.00	362.50				0.00	0.00	0.00	0.00
9740	GW	30.0	10.00		36.75	0.00	367.50				0.00	0.00	0.00	0.00
9750	GW	30.5	10.00		37.25	0.00	372.50				0.00	0.00	0.00	0.00
9760	GW	31.0	10.00		37.75	0.00	377.50				0.00	0.00	0.00	0.00
9770	GW	31.5	10.00		38.25	0.00	382.50				0.00	0.00	0.00	0.00
9780	GW	32.0	10.00		38.75	0.00	387.50				0.00	0.00	0.00	0.00
9790	GW	32.5	10.00		39.25	0.00	392.50				0.00	0.00	0.00	0.00
9800	GW	33.0	10.00		39.75	0.00	397.50				0.00	0.00	0.00	0.00
9810	GW	33.5	10.00		40.25	0.00	402.50				0.00	0.00	0.00	0.00
9820	GW	34.0	10.00		40.75	0.00	407.50				0.00	0.00	0.00	0.00
9830	GW	34.5	10.00		41.25	0.00	412.50				0.00	0.00	0.00	0.00
9840	GW	35.0	10.00		41.75	0.00	417.50				0.00	0.00	0.00	0.00
9850	GW	35.5	10.00		42.25	0.00	422.50				0.00	0.00	0.00	0.00
9860	GW	36.0	10.00		42.75	0.00	427.50				0.00	0.00	0.00	0.00
9870	GW	36.5	10.00		43.25	0.00	432.50				0.00	0.00	0.00	0.00
9880	GW	37.0	10.00		43.75	0.00	437.50				0.00	0.00	0.00	0.00
9890	GW	37.5	10.00		44.25	0.00	442.50				0.00	0.00	0.00	0.00
9900	GW	38.0	10.00		44.75	0.00	447.50				0.00	0.00	0.00	0.00
9910	GW	38.5	10.00		45.25	0.00	452.50				0.00	0.00	0.00	0.00
9920	GW	39.0	10.00		45.75	0.00	457.50				0.00	0.00	0.00	0.00
9930	GW	39.5	10.00		46.25	0.00	462.50				0.00	0.00	0.00	0.00
9940	GW	40.0	10.00		46.75	0.00	467.50				0.00	0.00	0.00	0.00
9950	GW	40.5	10.00		47.25	0.00	472.50				0.00	0.00	0.00	0.00
9960	GW	41.0	10.00		47.75	0.00	477.50				0.00	0.00	0.00	0.00
9970	GW	41.5	10.00		48.25	0.00	482.50				0.00	0.00	0.00	0.00
9980	GW	42.0	10.00		48.75	0.00	487.50				0.00	0.00	0.00	0.00
9990	GW	42.5	10.00		49.25	0.00	492.50				0.00	0.00	0.00	0.00
10000	GW	43.0	10.00		49.75	0.00	497.50				0.00	0.00	0.00	0.00
10010	GW	43.5	10.00		50.25	0.00	502.50				0.00	0.00	0.00	0.00
10020	GW	44.0	10.00		50.75	0.00	507.50				0.00	0.00	0.00	0.00
10030	GW	44.5	10.00		51.25	0.00	512.50				0.00	0.00	0.00	0.00
10040	GW	45.0	10.00		51.75	0.00	517.50				0.00	0.00	0.00	0.00
10050	GW	45.5	10.00		52.25	0.00	522.50				0.00	0.00	0.00	0.00
10060	GW	46.0	10.00		52.75	0.00	527.50				0.00	0.00	0.00	0.00
10070	GW	46.5	10.00		53.25	0.00	532.50				0.00	0.00	0.00	0.00
10080	GW	47.0	10.00		53.75	0.00	537.50				0.00	0.00	0.00	0.00
10090	GW	47.5	10.00		54.25	0.00	542.50				0.00	0.00	0.00	0.00
10100	GW	48.0	10.00		54.75	0.00	547.50				0.00	0.00	0.00	0.00
10110	GW	48.5	10.00		55.25	0.00	552.50				0.00	0.00	0.00	0.00
10120	GW	49.0	10.00		55.75	0.00	557.50				0.00	0.00	0.00	0.

Station/ Chainage (m)	Structure	Basin HL	Basin Length	Position	K-Area of GW	Surface Area of RE Wall (m ²)	Volume of GW (m ³)	RE Wall Actually Provided GW (m ²)	Add Qty GW in place replaced RE (m ²)	Understand Qty GW at Convert location (m ²)	Extra Qty GW at (m ²)	Backfill Qty for GW (m ³)	Extra Qty for GW (m ³)	Backfill Qty for REW (m ³)	Backfill Qty for REW (m ³)
82551.295	GW	4.0	1.79		11.75	0.00	17.16				25.35	17.38	0.00	0.00	0.00
82558.295	GW	4.0	7.50		11.75	0.00	99.38				166.83	71.12	0.00	0.00	0.00
82560	GW	4.0	1.21		11.75	0.00	15.87				22.59	11.43	0.00	0.00	0.00
82570	GW	3.5	10.00		12.75	0.00	137.50				178.20	72.17	0.00	0.00	0.00
82573.795	GW	3.5	3.79		12.75	0.00	48.39				67.63	37.39	0.00	0.00	0.00
82580	GW	3.5	6.71		12.75	0.00	79.11				110.57	44.76	0.00	0.00	0.00
82590	GW	3.5	10.00		12.75	0.00	137.50				178.20	72.17	0.00	0.00	0.00
82597.963	GW	4.5	7.96		16.75	0.00	133.36				178.20	72.17	0.00	0.00	0.00
82600	GW	4.5	2.04		16.75	0.00	34.12				48.93	23.72	0.00	0.00	0.00
82610	GW	4.0	10.00		13.75	0.00	133.50				195.77	94.87	0.00	0.00	0.00
82612.963	GW	3.5	2.96		12.75	0.00	37.38				52.80	21.38	0.00	0.00	0.00
82620	GW	2.0	7.04		6.63	0.00	32.55				52.80	21.38	0.00	0.00	0.00
82626.463	GW	1.0	6.46		2.75	0.09	17.77				18.06	7.46	0.00	0.00	0.00
82630	GW	1.0	3.54		2.75	0.00	9.73				9.88	4.00	0.00	0.00	0.00
82640	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00	0.00
82650	GW	5.5	10.00		21.25	0.00	212.50				309.58	126.45	0.00	0.00	0.00
82656.276	RE	7.0	6.28		0.00	0.00		0.00	0.00	0.00			0.00	0.00	0.00
82660	RE	7.5	3.72		0.00	0.00		0.00	0.00	0.00			0.00	0.00	0.00
82670	RE	6.5	10.00		0.00	0.00		0.00	0.00	0.00			0.00	0.00	0.00
82680	GW	2.0	10.00		4.63	0.00	46.35				76.54	35.36	0.00	0.00	0.00
82690	FWA W	0.0	0.00		0.00	0.00							0.00	0.00	0.00
82692.276	FWA W	0.0	0.00		0.00	0.00							0.00	0.00	0.00
82500	GW	1.0	7.77		2.75	0.00	27.24				0.00	0.00	0.00	0.00	0.00
82510	GW	3.5	10.00		12.75	0.00	137.50				178.20	72.17	0.00	0.00	0.00
82520	GW	6.0	10.00		21.25	0.00	212.50				329.54	214.88	0.00	0.00	0.00
82532.315	RE	7.5	10.00		0.00	0.00		9.00	0.00	0.00			0.00	0.00	0.00
82540	RE	7.5	7.68		0.00	0.00		9.00	0.00	0.00			0.00	0.00	0.00
82550	GW	4.0	10.00		13.75	0.00	133.50				195.77	94.87	0.00	0.00	0.00
82562.315	GW	1.0	2.32		2.75	0.00	27.50				11.55	2.67	0.00	0.00	0.00
82570	FWA W	0.0	0.00		0.00	0.00							0.00	0.00	0.00
82580	FWA W	0.0	0.00		0.00	0.00							0.00	0.00	0.00
82590	FWA W	0.0	0.00		0.00	0.00							0.00	0.00	0.00
82600	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00	0.00
82607.261	GW	2.5	2.26		9.25	0.00	20.91				27.94	10.44	0.00	0.00	0.00
82610	GW	4.0	7.24		13.75	0.00	102.54				151.51	73.16	0.00	0.00	0.00
82615.763	GW	4.0	5.76		13.75	0.00	76.33				112.79	54.63	0.00	0.00	0.00
82620	GW	4.0	4.24		13.75	0.00	56.17				80.99	40.20	0.00	0.00	0.00
82630	GW	3.5	10.00		12.75	0.00	127.50				178.20	72.17	0.00	0.00	0.00
82630.763	GW	3.5	0.76		12.75	0.00	9.70				13.56	5.69	0.00	0.00	0.00
82640	GW	2.5	9.24		9.25	0.00	85.46				114.18	42.67	0.00	0.00	0.00
82650	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00	0.00
82660	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00	0.00
82670	GW	2.5	10.00		9.25	0.00	92.50				123.58	46.19	0.00	0.00	0.00
82680	RE	8.0	10.00		0.00	0.00		0.00	0.00	0.00			0.00	0.00	0.00
82690	GW	1.0	10.00		2.75	0.00	27.50			0.00	27.95	11.55	0.00	0.00	0.00
82700	FWA W	0.0	0.00		0.00	0.00							0.00	0.00	0.00
82710	FWA W	0.0	0.00		0.00	0.00							0.00	0.00	0.00
82720	FWA W	0.0	0.00		0.00	0.00							0.00	0.00	0.00
82730	FWA W	0.0	0.00		0.00	0.00							0.00	0.00	0.00
82744.576	GW	2.5	4.48		4.13	0.00	41.30				63.85	25.98	0.00	0.00	0.00
82750	GW	4.5	5.32		16.75	0.00	169.50				240.20	116.47	0.00	0.00	0.00
82760	GW	3.5	1.76		12.75	0.00	39.43				98.44	39.87	0.00	0.00	0.00
82771.793	GW	2.5	7.79		9.25	0.00	72.09				96.31	35.93	0.00	0.00	0.00
82780	GW	2.0	2.21		4.63	0.00	10.21				16.89	7.80	0.00	0.00	0.00
82790	GW	1.5	10.00		4.13	0.00	41.30				63.85	25.98	0.00	0.00	0.00
82799.9	GW	2.5	9.90		9.25	0.00	91.57				122.34	45.73	0.00	0.00	0.00
82800	GW	2.0	0.10		4.63	0.00	0.46				0.77	0.35	0.00	0.00	0.00

Summary of Unit Rates for Meirangkong - Tamlu - Mon Road

Item No	Description	Unit	Rate as per MORTH Data Book (Rs)	Item No. in Rate Analysis
1	2	3	4	5
BILL NO. 8: DRAINAGE AND PROTECTIVE WORKS				
8.01	Lined Random Rubble Masonry Drain on Hill Side, Embankment Toe and Hill Crest Complete as per drawings and Technical Specification Section 300 and 1400 and as directed by the Engineer.	m	4389	
8.02	Covered RCC Rectangular Drain including Reinforcement complete as per drawing and Technical Specification Sections 300, 1000, 1400, 1500, 1600, 1700 and as directed by Engineer	m	18804	
8.03	Subsurface Drainage complete as per drawings and Technical Specification Section 309			
a	100 mm diameter PVC Perforated Closed Jointed wrapped with Geo-synthetic as per drawings and Technical Specification	m	463	
b	Aggregate Drain below Hill Side Drain as per drawings and Technical Specification	m	379	
8.04	RCC Retaining wall, Gabion Wall Excavation for RCC retaining wall, Gabion wall as per drawings and Technical Specification, in			
i	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	235	(3.32+3.33)/2
ii	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	393	3.34
iii	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	385	3.9
iv	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	515	3.8 A
b	Back filling behind Earth Retaining Structures Complete as per drawings and Technical Specification Clause 305.	cum	3670	13.9 A
c	Boulder / Concrete Block Piling on Cut / Embankment Slopes complete as per drawings and Technical Specification Section 2500.	cum	4283	15.4
8.05	Gabion Structure for Retaining Earth with segments of mechanically woven (double twist) wire crates filled with boulders including non-woven geotextile as filter media complete as per drawing and Technical Specification Section 700 and 2500	cum	6325	As per Market
8.06	Seeding and Mulching of Cut Slopes / Embankment Slopes complete as per drawings and Technical Specification Clause 308	sqm	995	3.23
8.07	Geo-synthetic Mat for Erosion Control complete as per drawings and Technical Specification clause 706	sqm	1860	As per Market
8.08	Soil Nailing complete as per drawings and Technical Specification Section 3200	m	4893	As per Market
8.09	Temporary Facing complete as per drawings and Technical Specification 3200.	sqm	3045	As per Market
8.10	Reinforced Soil Wall complete as per drawing and Technical Specification Section 3100.			
a	Face panels as per Technical Specification Section 3105 including soil reinforcing element, foundation pad, coping beam, all accessories, consumables and components of drainage system (filter media, drainage layer, drain pipe, catch pit etc.), including ground improvement complete.	sqm	7142	7.5 (I) A (type-5) + 7.5(II) + 12.8 B*0.20
b	Excavation for RE wall as per drawings and Technical Specification, in			
i	All types of soils and Ordinary Rocks as per Technical specification Clause 304.	cum	235	(3.32+3.33)/2
ii	Hard Rock (requiring blasting) as per Technical Specification Clause 302 and 304.	cum	393	3.34
iii	Hard Rock (using controlled blasting) as per Technical Specification Clause 302 and 304.	cum	385	3.9
iv	Hard Rock (blasting prohibited) as per Technical Specification Clause 304	cum	515	3.8 A
c	Back filling behind RE Wall Complete as per drawings and Technical Specification Clause 305.	cum	3670	13.9 A
8.11	Construction of Parapet Wall in RR Masonry (CM 1:3) as per Drawings and Technical Specification 1400	cum	10550	13.4 A

Quantity Calculation for Traffic Signs

Annexure -9.01

SL.NO.	NAME	CODE	LOCATION OF CAUTIONARY SIGNS															KM15-KM16	
			0KM-1KM	1KM-2KM	2KM-3KM	3KM-4KM	4KM-5KM	5KM-6KM	6KM-7KM	7KM-8KM	8KM-9KM	9KM-10KM	10KM-11KM	11KM-12KM	12KM-13KM	13KM-14KM	14KM-15KM		
	STOP	14.01	3	1	2	2		1				1	1	4	1	1		1	
	GIVE WAY	14.02	1																
	GIVE WAY TO BUSES EXISTING THE BUS BAY	14.03																	
	NO STOPPING OR STANDING	14.28																	
	NO PARKING	14.29																	
	MAXIMUM SPEED LIMIT	14.37	2				1	1		1		1	2						
	COMPULSORY KEEP LEFT	14.48																	
	COMPULSORY KEEP RIGHT	14.49	3										3						
	COMPULSORY SOUND HORN	14.57	2	2		2	2	2	2	2	2	2	2	2		2	1	1	
	LEFT HAND CURVE	15.01																	
	RIGHT HAND CURVE	15.02																	
	RIGHT HAIRPIN BEND	15.03					2												
	LEFT HAIRPIN BEND	15.04					2		1	1	1	2							
	RIGHT REVERSE BEND	15.05	2	3		1			2	2	2	2				1	1	2	
	LEFT REVERSE BEND	15.06	2	3		1			1	1	1	2				1	1	2	
	SERIES OF BEND	15.07	2	2	2	2	4		2	2	1	4	2			2	2	2	
	SIDE ROAD RIGHT	15.09	3	1	2	2						1		4	1	1	1	1	
	SIDE ROAD LEFT	15.10	3	1	2	2						1		4	1	1	1	1	
	Y-INTERSECTION LEFT	15.11																	
	Y-INTERSECTION RIGHT	15.12																	
	Y-INTERSECTION	15.13																	
	X-ROAD	15.14																	
	ROUNDABOUT	15.15																	
	T-INTERSECTION	15.17																	
	T-INT. MAJOR ROAD	15.18																	
	STAGGERED INTERSECTION RIGHT	15.20																	
	STAGGERED INTERSECTION LEFT	15.21																	
	NARROW BRIDGE AHEAD	15.25																	
	STEEP ASCENT	15.26																	
	STEEP DESCENT	15.27												1					
	PEDESTRIAN CROSSING	15.33	2		2		1	1	1	1	1			1					
	SCHOOL A-HEAD	15.34																	
	BUILT UP AREA	15.35	2	2	2									2					
	PLAYGROUND AHEAD	15.58												2					
	BARRIER	15.60												2					
	FALLING ROCKS	15.65																	
	SINGLE CHEVRON (SMALL)	15.72	32	35	37	44	32	30	25	29	31	30	38	24	33	18	30	50	
	DOUBLE CHEVRON	15.74																	
	OBJECT MARKER(LEFT)	15.76	4	5	7	2	5	1	3	3	3	2	4	1	4	2	5	7	
	OBJECT MARKER(RIGHT)	15.77	4	5	7	2	5	1	3	3	3	2	4	1	4	2	5	7	
	TWO WAY HAZARD MARKER	15.78																	
	MAP TYPE ADVANCE DIRECTION SIGN	16.02	3																
	REASSURANCE SIGN	16.05																	
	FLAG TYPE DIRECTION SIGN	16.04	2		4	4									2	2			
	PLACE IDENTIFICATION SIGN	16.06	2	2	2														
	TRUCK LAY- BY	16.07																	
	WEIGH BRIDGE AHEAD	16.09																	
	GANTORY BOARD	16.10	1																
	TOILET	17.05																	
	FILLING STATION	17.06																	
	HOSPITAL	17.07																	
	POLICE STATION	17.13																	
	PICNIC SITE	17.14																	
	HOME ZONE	17.21																	
	BUSTOP	17.35	2	2	2														
	STATE HIGHWAY ROUTE MARKER SIGN	20.01	5		2	2													
	NATIONAL HIGHWAY ROUTE MARKER SIGN	20.02																	
	TOTAL NO OF SIGNS			80	64	73	68	55	40	37	44	49	48	59	63	50	31	48	73